



Swiss Institute of
Bioinformatics

First Steps with R in Life Sciences: Introduction

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ELIXIR Code of conduct



A place to feel respected
A place to feel safe



General Information

Course page: <https://github.com/sib-swiss/first-steps-with-R-training>



- Slides
- Data sets
- Exercises
- Solutions

Optional exam, 0.5 ECTS value

Asking questions - Communication

- Raise your hand anytime



- Done with an exercise?



Introducing ourselves



Who are you?

Your name, and where you work or study



What do you work on?

Your field, and the kind of data you handle



Why R?

*What you would like to be able to do after
this course*

What type of computer OS are you using?



Windows



macOS



Unix / Linux

Ubuntu, CentOS, ...

Why we ask: R is the same everywhere, but shortcuts (Ctrl vs Cmd) and file paths (C:/Users/... vs /Users/...) differ a little.

Do you code in other languages?



None

R will be your first one, great place to start



Yes, but only notions

you have seen a loop or two



Yes, and I am proficient

you will mostly learn R's quirks

For example: Python, bash, MATLAB, Java, C, SQL, Perl...



What is your experience with R?



Never tried

I have never used R



A few commands

I ran some commands or did a few exercises



A bit for real

I used R a little for work or a personal project



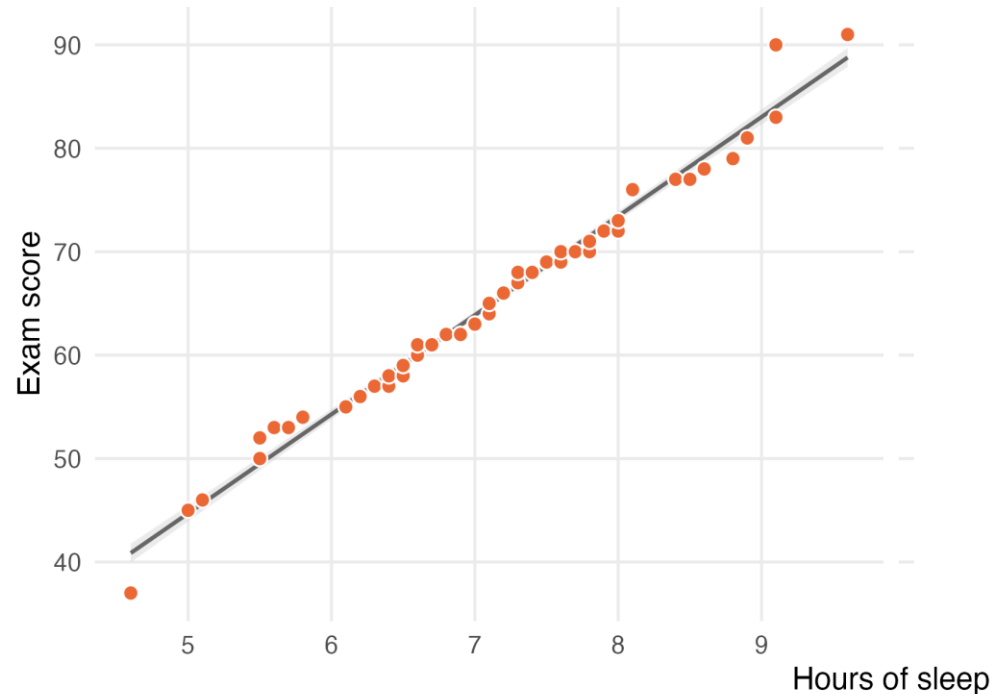
Extensively

...wait, why am I even here?

Every level is welcome: the exercises have extra tasks for the fast ones.

Why learn R?

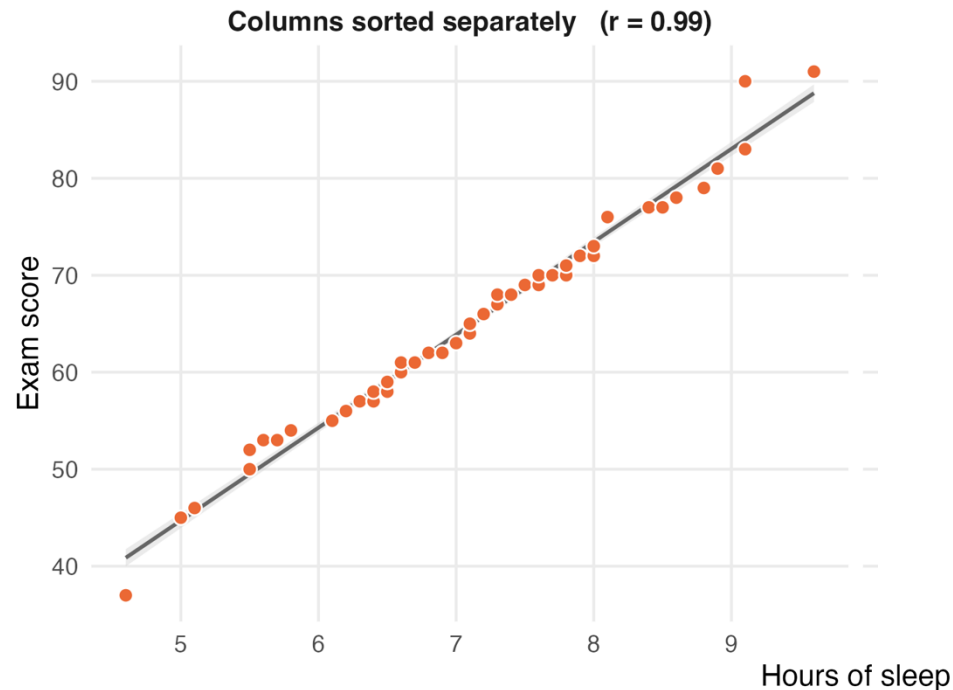
- Suppose you have made some nice plots to show your collaborators



A tip found online: as a first step, tidy up the columns of your data frame with the `sort()` function

Why learn R?

- Suppose you have made some nice plots to show your collaborators

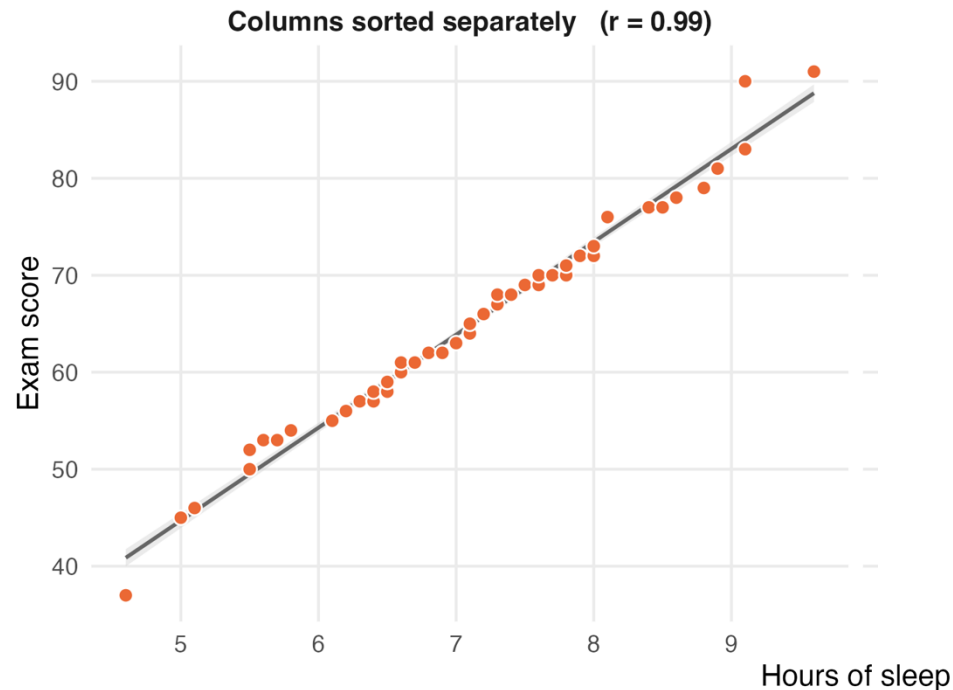


A tip found online: as a first step, tidy up the columns of your data frame with the `sort()` function

```
tidy <- students  
tidy$sleep <- sort(tidy$sleep) # <- looks harmless...  
tidy$score <- sort(tidy$score) # <- ...but is it?
```

Why learn R?

- Suppose you have made some nice plots to show your collaborators



```
> head(students) # original: student 1 slept X hours and scored Y
  id sleep score
1  1  6.3   70
2  2  6.7   59
3  3  8.9   61
4  4  7.1   53
5  5  7.2   52
6  6  9.1   69

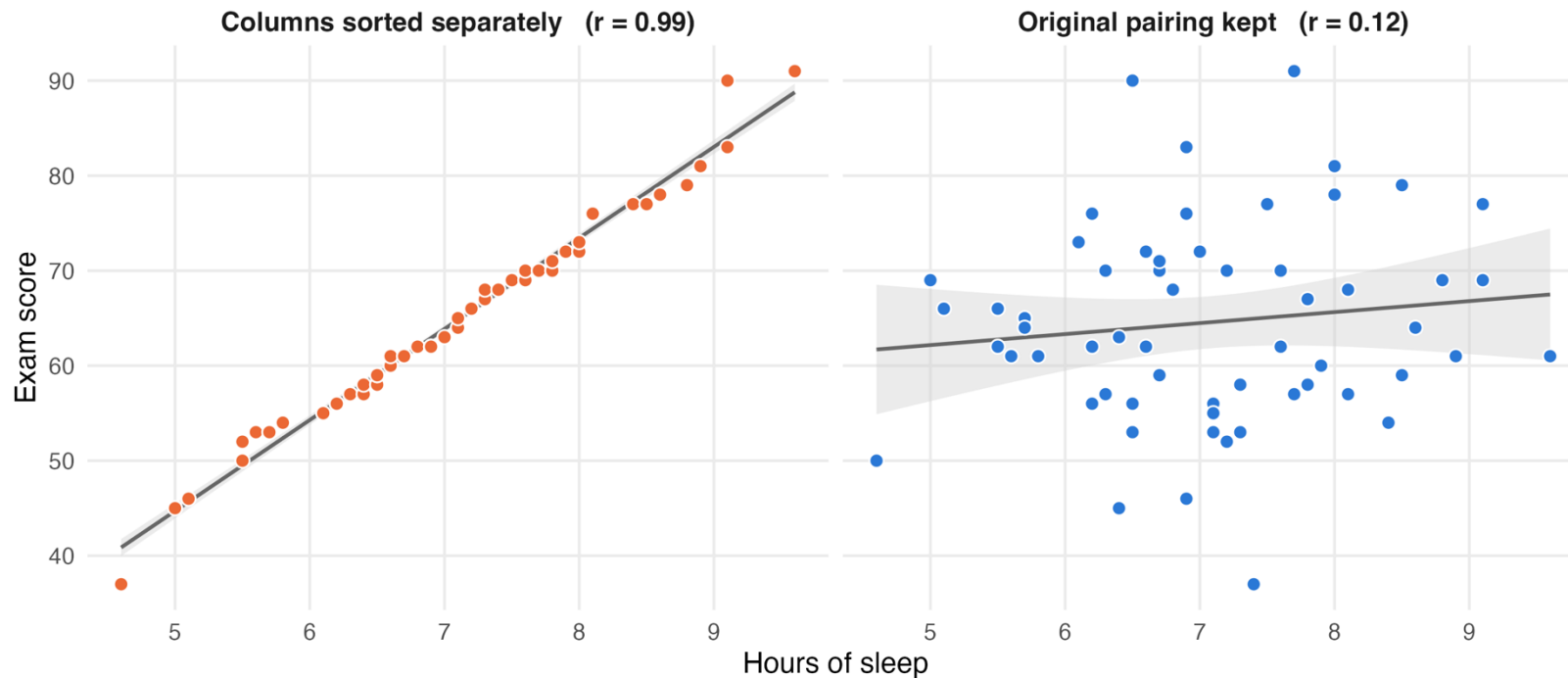
> head(tidy) # "tidied": the smallest sleep is now paired with the smallest score
  id sleep score
1  1  4.6   37
2  2  5.0   45
3  3  5.1   46
4  4  5.5   50
5  5  5.5   52
6  6  5.6   53
```

Why learn R?

- Better make sure you know what your script is doing!

Same numbers, one line of code apart

`sort()` on each column broke the link between a student's sleep and their score



Course Content:
R is vast, it can't be learned overnight.

This course will give a **basic understanding and concepts of R.**

This course is the first step in your  journey

Course Outline

Day 1

1. Getting Familiar with **R** and **Rstudio**
2. Getting started with the **R syntax and Objects**
3. **Packages and versioning**
4. **R environnement**
5. Formatting your data
6. **Importing/exporting data** with R (working with files)

Day 2

1. Building **Graphics** in R (basic plotting)
2. R functions
3. Data analysis in R
4. Introduction to R markdown
5. High-definition figures export
6. Statistics with R: an introduction

Getting Familiar with **R** and **Rstudio**

RGui (R Graphical user interface)

Installed at the same time as the programming language

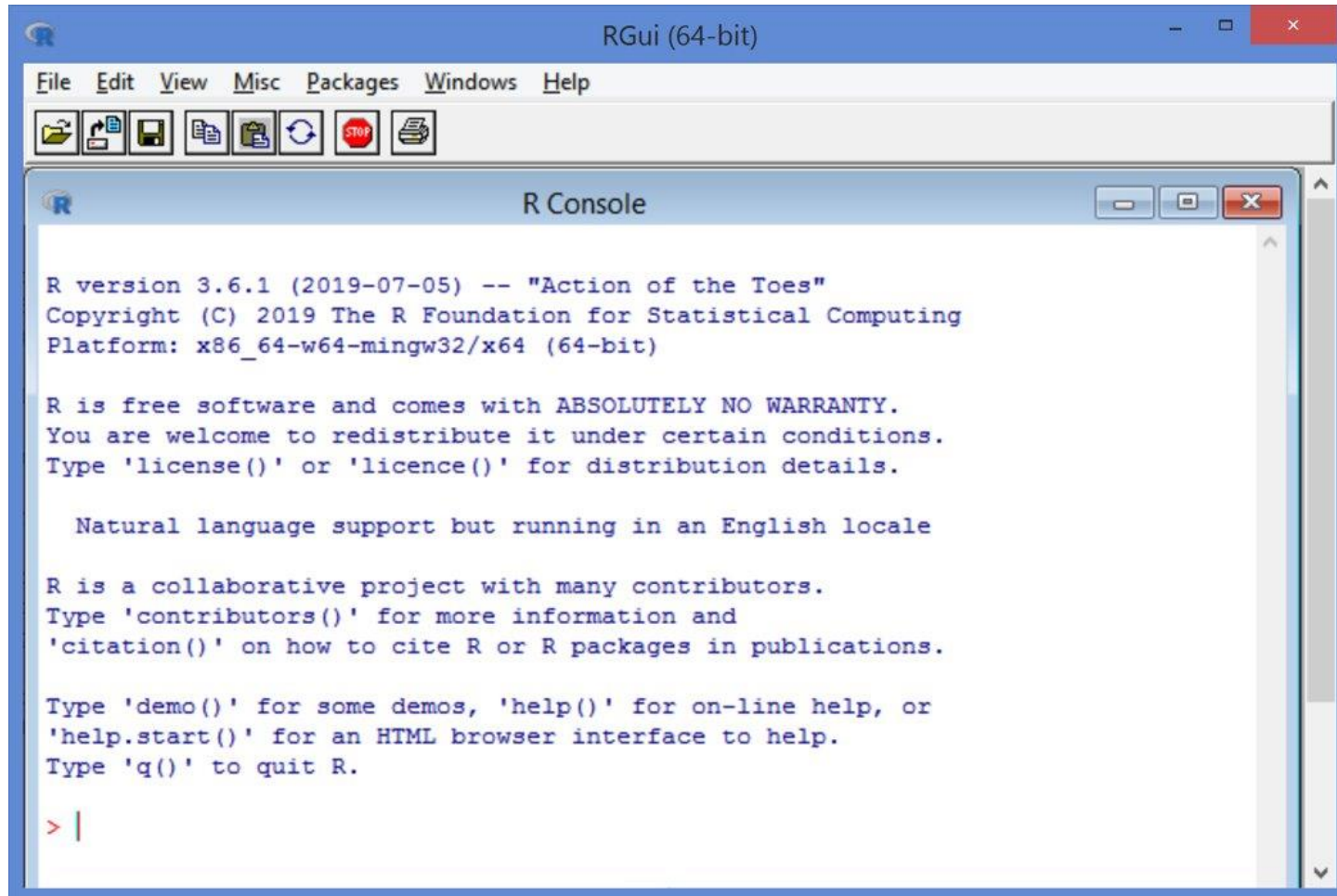


Current version

R 4.6.1

June 2026

This screenshot runs R 3.6.1 from 2019. In R years, that is roughly the Jurassic. Yours will look the same, just younger.

A screenshot of the RGui (64-bit) window. The window has a blue title bar and a menu bar with options: File, Edit, View, Misc, Packages, Windows, and Help. Below the menu bar is a toolbar with icons for file operations and execution. The main area is the R Console, which displays the following text:

```
R version 3.6.1 (2019-07-05) -- "Action of the Toes"
Copyright (C) 2019 The R Foundation for Statistical Computing
Platform: x86_64-w64-mingw32/x64 (64-bit)

R is free software and comes with ABSOLUTELY NO WARRANTY.
You are welcome to redistribute it under certain conditions.
Type 'license()' or 'licence()' for distribution details.

Natural language support but running in an English locale

R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

> |
```

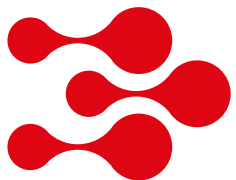
R combined with Rstudio



Integrated Development Enviroment (IDE) designed to help with R

It includes:

- A **console**
- A **syntax highlighting editor**
- Tools for viewing the **workspace** and **history**
- A file explorer, package explorer, **plot** and **help** display areas



We suggest Rstudio as the preferred option when working with R

Rstudio Interface

The screenshot displays the RStudio interface with the following components:

- Editor:** The central pane shows an R script with the following code:

```
1 ##### My first script #####
2 ##### October 2017 #####
3
4
5 # list workspace
6 ls()
7
8 # Reset R's brain
9 rm(list=ls())
10
11 # check wd
12 getwd()
13
14 #set wd
15 setwd("~/Users/dmarek/EducationSIB/Courses_2016/First_Steps_R_June2016/R_intro_course")
16
17 # confirm wd
18 getwd()
19
20 #load packages if needed (to do every time you launch your R session)
21 #library("boot")
22 #library("lattice")
23
24 #####
25
26 ##### October 2017 #####
```
- Environment:** The top right pane shows the workspace and history. It lists the following data objects:

Data	Values
mice_data	50 obs. of 3 variables
mice_weight_HFD	29 obs. of 3 variables
mids	num [1:2, 1] 0.7 1.9
mean_weight_diet	num [1:2(1d)] 28.7 37.1
mean_weight_genotype	num [1:2(1d)] 33.7 33.4
n_weight_diet	int [1:2(1d)] 21 29
n_weight_genotype	int [1:2(1d)] 24 26
sd_weight_diet	num [1:2(1d)] 2.61 5
sd_weight_genotype	num [1:2(1d)] 4.69 6.92
- Console:** The bottom left pane shows the R console output, which includes the R license text and the message: `[Workspace loaded from ~/TrainingSIB/Courses_2017/First_Steps_R_Oct2017/.RData]`
- File Explorer:** The bottom right pane shows the file explorer, displaying a list of files in the `~/TrainingSIB/Courses_2017/First_Steps_R_Oct2017/course_datasets` directory.

Workspace, history

Editor

Console, terminal

File explorer, plots, packages, help

From [Posit PBC](#), the creators of RStudio

The data science IDE. Human-driven, AI-assisted.

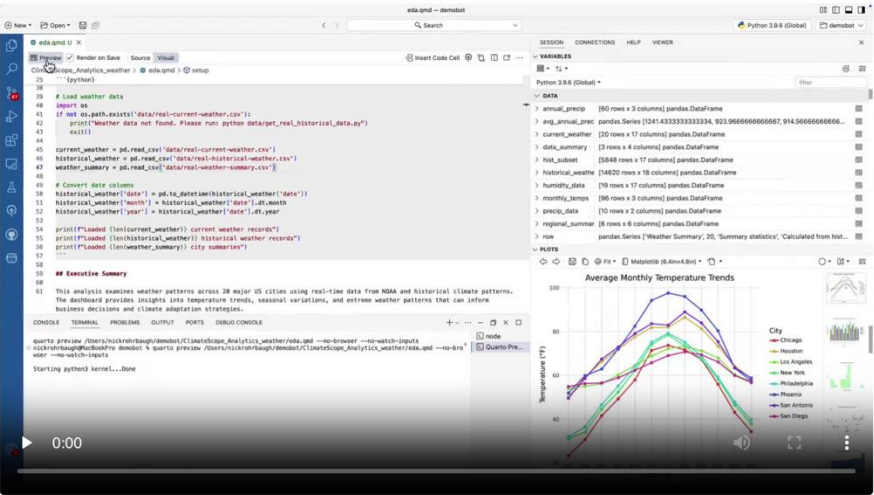
Move from question to insight to analysis, model, or application faster than ever before. Positron is the free IDE made specifically for data science and analysis with Python and R.

Download

Explore Features

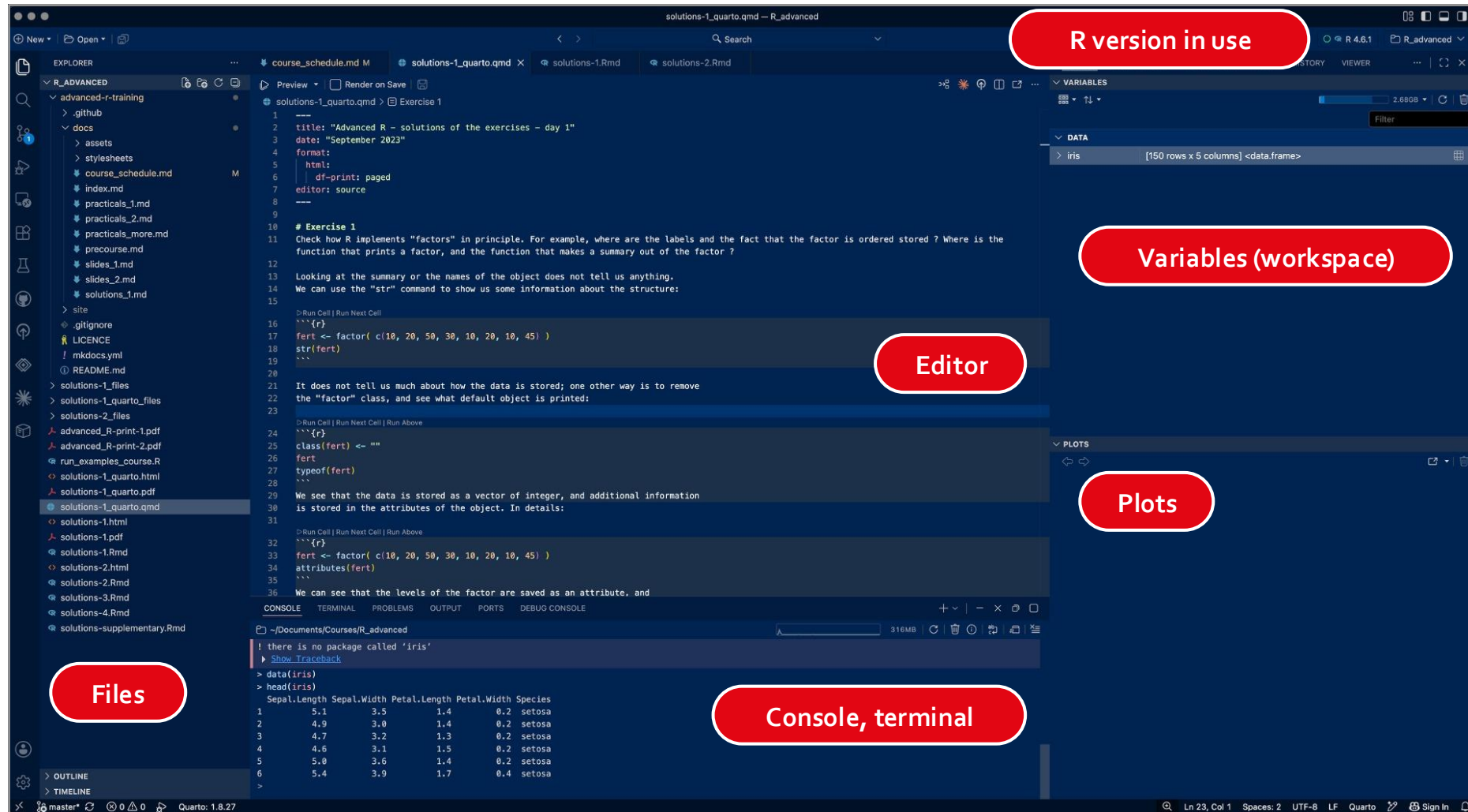
Get Updates

Positron by Posit



Positron™ is licensed under the [Elastic License 2.0](#), a source-available license. [Read more](#) about what this license means and our decision to use it.

Positron interface



Same building blocks as RStudio, arranged a bit differently. Help, History and Viewer sit in tabs next to Session.

Positron: an alternative to RStudio

A free, next-generation IDE released by **Posit** (the company behind RStudio). Stable since summer 2025.

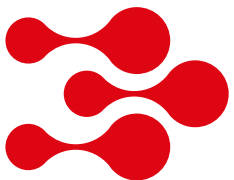
	RStudio	Positron
Languages	Built for R (Python possible)	R and Python treated as equals
Based on	Its own interface, 14+ years old	Code OSS, the open-source core of VS Code
Interface	4 panes: editor, console, environment, files/plots/help	Very similar panes, plus a data explorer and extensions
Robustness	A crash in R can freeze the whole IDE	R runs separately: a crash does not kill the IDE
Projects	.Rproj files	Plain folders (workspaces)

In this course we use RStudio. Everything you learn here (console, scripts, working directory, packages) transfers directly to Positron. Feel free to use Positron though.

Creating an R project

Rstudio allows the organization of your work in **projects**

- **File > New Project** or click on "Project" in the upper-right corner
- Choose **New directory**, then New Project, give an appropriate name and location
- OR choose **Existing Directory**
- This creates an **.Rproj file** in the directory



The project directory automatically becomes the "working directory"

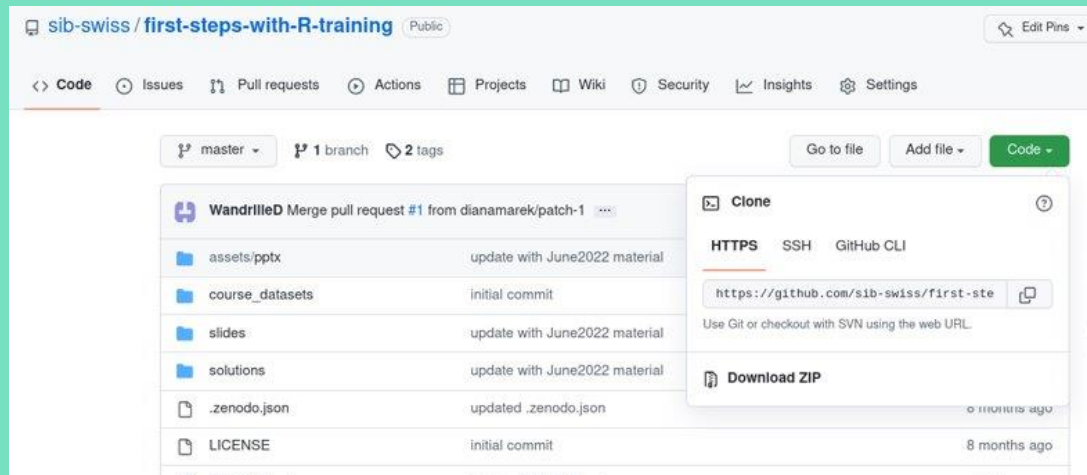
Let's practice - 1

1. **Outside Rstudio : prepare the course data for the exercises**

Download the course material from :

<https://github.com/sib-swiss/first-steps-with-R-training>

Either use git clone OR click **Download ZIP**



and **UNZIP** and then move the folder where you want it

2. **Inside Rstudio: project setup**

In Rstudio, create a new project set in the folder of the course material you just recovered.

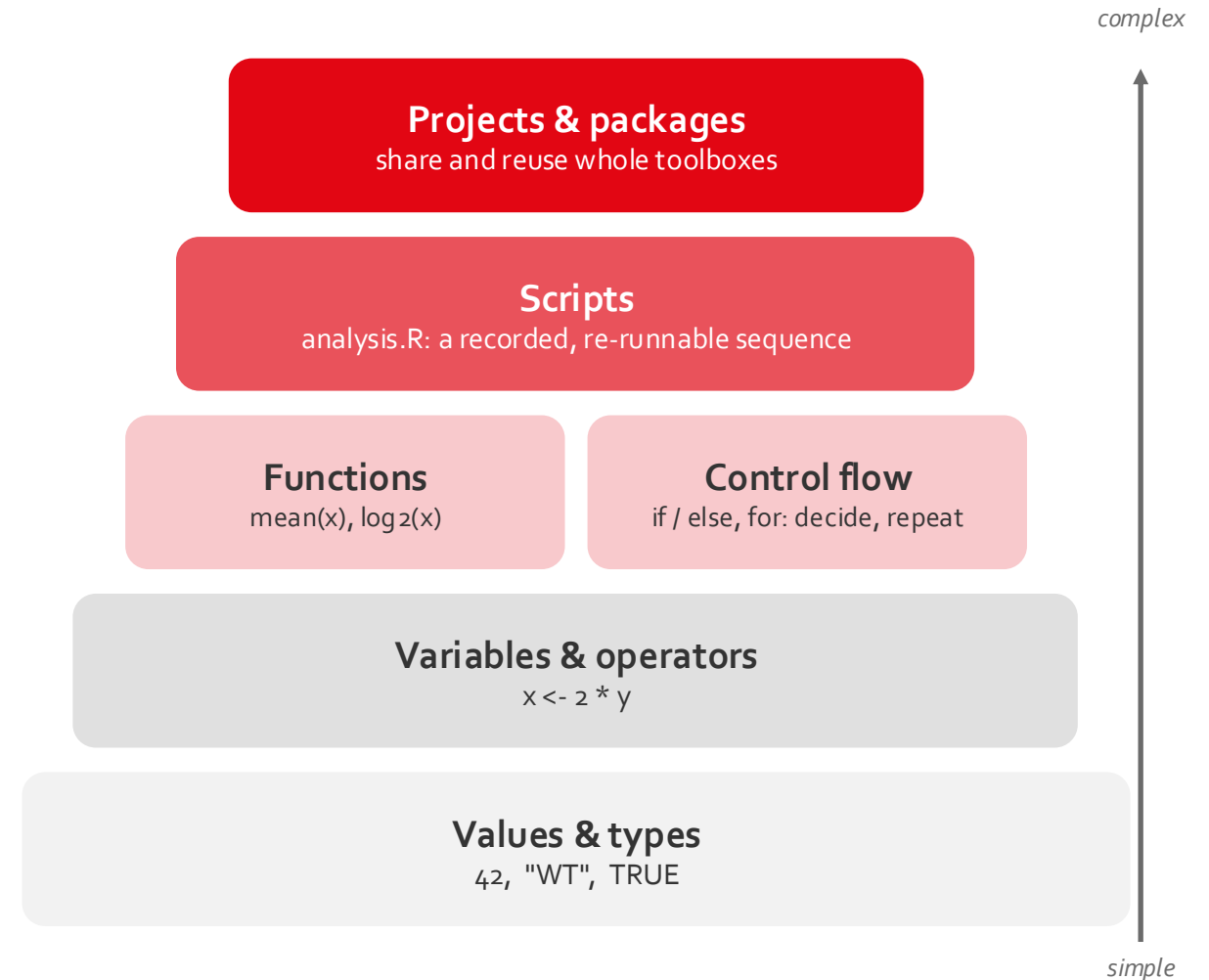
What is a programming language?

A **programming language** is a set of words and rules to give **precise, unambiguous instructions** to a computer.

R is an **interpreted** language: each command is read and executed right away by the R interpreter.

The computer does exactly what you write, not what you meant.

How do we write complex programs? By stacking simple building blocks: each level reuses the ones below it, so you never have to think about everything at once.



Today we climb the first steps of this ladder.

Console: the command line

~/TrainingSIB/Courses_2017/First_Steps_R_Oct2017/ ↗

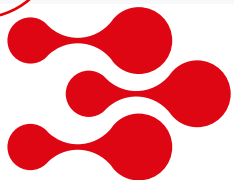
R est un logiciel libre livré sans AUCUNE GARANTIE.
Vous pouvez le redistribuer sous certaines conditions.
Tapez 'license()' ou 'licence()' pour plus de détails.

R est un projet collaboratif avec de nombreux contributeurs.
Tapez 'contributors()' pour plus d'information et
'citation()' pour la façon de le citer dans les publications.

Tapez 'demo()' pour des démonstrations, 'help()' pour l'aide
en ligne ou 'help.start()' pour obtenir l'aide au format HTML.
Tapez 'q()' pour quitter R.

[Workspace loaded from ~/TrainingSIB/Courses_2017/First_Steps_R_Oct2017/.RData]

> |



The ">" indicates R is ready for a new command

R key concepts: a first glimpse

You just met four building blocks of R. A first glimpse only: we come back to each of them in detail later.

```
1 + 1
```

Operator

does arithmetic or comparisons

+ - * / ^

```
abs(-11)
```

Function

ready-made code: name + parentheses

returns 11

```
x <- 128.5
```

Variable

stores a value under a name

x now contains 128.5

```
# my note
```

Comment

ignored by R, written for humans

explain your code!

Try it out:

Type the following at the command prompt:

Simple calculation:

```
1+1
```

Pre-defined function:

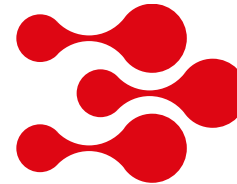
```
abs (-11)
```

Assigning a value to a variable

```
x <- 128.5
```

Displaying a variable's content

```
x
```



After each command,
Hit the return key to execute it

R key concepts: the working directory

The **working directory** is the folder where R reads and writes files by default.

In an RStudio project, it is automatically the project folder, so you rarely need to change it.

Where am I?

```
getwd()
[1] "C:/Users/Leo/Rcourse"
```

Move elsewhere (rarely needed)

```
setwd("D:/cool_R_project/")
```

Your computer

```
C:/Users/Leo/
├─ Documents/
└─ Rcourse/
    ├─ Rcourse.Rproj
    ├─ course_datasets/
    │   └─ mice_data.csv
    └─ ex1.R
```

← you are here

So R finds "course_datasets/mice_data.csv" without the full path.

R scripts

A script is a **normal text file which contains commands** meant to be executed one after the other.

Write your code into a script and save it

- to document your analysis as you do it
 - to be able to re-use that code later and create variations
 - for easy execution
-
- In Rstudio : **File > New File > Rscript** or **Ctrl+Shift+N**
 - **Save often!** (File > Save or Ctrl+S)

Rstudio Interface

The screenshot displays the RStudio interface with the following components:

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```
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2 ##### October 2017 #####
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4
5 # list workspace
6 ls()
7
8 # Reset R's brain
9 rm(list=ls())
10
11 # check wd
12 getwd()
13
14 #set wd
15 setwd("~/Users/dmarek/EducationSIB/Courses_2016/First_Steps_R_June2016/R_intro_course")
16
17 # confirm wd
18 getwd()
19
20 #load packages if needed (to do every time you launch your R session)
21 #library("boot")
22 #library("lattice")
23
24 2:26 October 2017
```
- Console, terminal:** The bottom-left pane shows the R startup message and workspace information:

```
R est un logiciel libre livré sans AUCUNE GARANTIE.
Vous pouvez le redistribuer sous certaines conditions.
Tapez 'license()' ou 'licence()' pour plus de détails.

R est un projet collaboratif avec de nombreux contributeurs.
Tapez 'contributors()' pour plus d'information et
'citation()' pour la façon de le citer dans les publications.

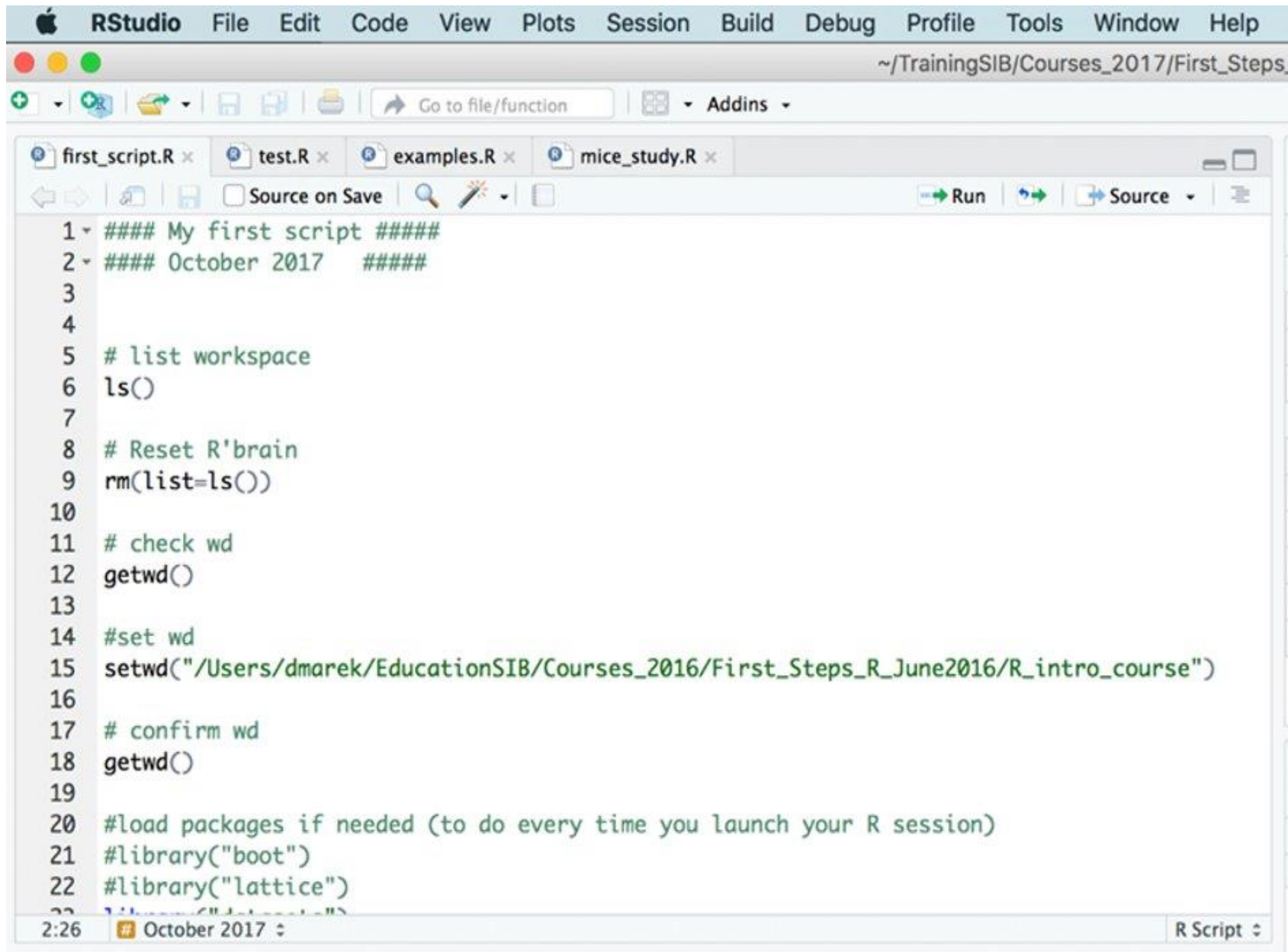
Tapez 'demo()' pour des démonstrations, 'help()' pour l'aide
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Tapez 'q()' pour quitter R.

[Workspace loaded from ~/TrainingSIB/Courses_2017/First_Steps_R_Oct2017/.RData]
```
- Workspace, history:** The top-right pane shows the Environment tab with the following data:

Variable	Value
mice_data	50 obs. of 3 variables
mice_weight_HFD	29 obs. of 3 variables
mids	num [1:2, 1] 0.7 1.9
- File explorer, plots, packages, help:** The bottom-right pane shows the Files tab with a list of files in the directory `~/TrainingSIB/Courses_2017/First_Steps_R_Oct2017/course_datasets`:

Name	Size	Modified
..		
class.csv	402 B	Mar 6, 2016, 6:33 PM
etubiol.csv	2.4 KB	Mar 7, 2016, 12:13 PM
mammals_survey.csv	1.2 MB	Jan 30, 2017, 7:41 AM
melanoma_data.txt	5.9 KB	Jan 30, 2017, 7:41 AM
mice_data.csv	632 B	Feb 28, 2016, 4:15 PM
my_data_frame.csv	262 B	Mar 2, 2015, 4:49 PM
pigs.csv	206 B	Mar 6, 2016, 6:32 PM
smoker.csv	4.6 KB	Feb 26, 2015, 12:23 PM
snpc.csv	618 B	Mar 6, 2016, 9:19 PM

Editor: writing code in a file



The screenshot shows the RStudio application window. The title bar includes the Apple logo and the text 'RStudio'. The menu bar contains 'File', 'Edit', 'Code', 'View', 'Plots', 'Session', 'Build', 'Debug', 'Profile', 'Tools', 'Window', and 'Help'. The status bar at the bottom shows the time '2:26', a file icon, the text '# October 2017 :', and 'R Script :'. The editor pane displays the following R code:

```
1 ##### My first script #####
2 ##### October 2017 #####
3
4
5 # list workspace
6 ls()
7
8 # Reset R'brain
9 rm(list=ls())
10
11 # check wd
12 getwd()
13
14 #set wd
15 setwd("/Users/dmarek/EducationSIB/Courses_2016/First_Steps_R_June2016/R_intro_course")
16
17 # confirm wd
18 getwd()
19
20 #load packages if needed (to do every time you launch your R session)
21 #library("boot")
22 #library("lattice")
23 #library("data.table")
```

Console or script: what is the difference?

R console

in a terminal or RGui

How: Type a command, press Enter: it runs immediately. Tab completion works; plots and help open in separate windows.

Code saved? No, only a command history.

Best for: Quick checks; working on a remote server (HPC).

Console in RStudio

bottom-left pane

How: Same R, same behaviour, but everything in one window: richer autocompletion, plus Environment, Plots and Help panes.

Code saved? No, only the History pane.

Best for: Trying things out; inspecting objects.

Script + console

editor pane, .R file

How: Write code in a text file, then send it to the console with Ctrl+Enter or Run.

Code saved? Yes: the file is your record.

Best for: Your real analysis: reproducible, commented, shareable.



There is only one R: the console is where code runs, the script is where your work is kept.

Sending code from the editor to the console

- Ctrl+Enter (Windows and Linux)
- Cmd+return (Mac)
- click the "Run" button

By default, the current line is executed,
if something is selected, then this is executed (e.g., part of a line, or multiple lines)

Most of your code should be in scripts

Code in scripts is "just text",
it gets executed when you send it to the console

It is possible to run an individual code line,
a block of code
or an entire script at once

Rstudio Interface

The screenshot displays the RStudio interface with the following components:

- Editor:** The central pane showing an R script with the following code:

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21 #library("boot")
22 #library("lattice")
23
24 2:26 October 2017
```

Environment, History, Connections: The top-right pane, highlighted with a red box, showing the current workspace and history. It lists the following objects:

Object	Details
mice_data	50 obs. of 3 variables
mice_weight_HFD	29 obs. of 3 variables
mids	num [1:2, 1] 0.7 1.9
mean_weight_diet	num [1:2(1d)] 28.7 37.1
mean_weight_genotype	num [1:2(1d)] 33.7 33.4
n_weight_diet	int [1:2(1d)] 21 29
n_weight_genotype	int [1:2(1d)] 24 26
sd_weight_diet	num [1:2(1d)] 2.61 5
sd_weight_genotype	num [1:2(1d)] 4.69 6.92

Console, terminal: The bottom-left pane showing the R console output. It displays the R license text and the workspace loaded from the specified path.

Files, Plots, Packages, Help, Viewer: The bottom-right pane, showing the file explorer. It lists the following files:

Name	Size	Modified
..		
class.csv	402 B	Mar 6, 2016, 6:33 PM
etubiol.csv	2.4 KB	Mar 7, 2016, 12:13 PM
mammals_survey.csv	1.2 MB	Jan 30, 2017, 7:41 AM
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pigs.csv	206 B	Mar 6, 2016, 6:32 PM
smoker.csv	4.6 KB	Feb 26, 2015, 12:23 PM
snps.csv	618 B	Mar 6, 2016, 9:19 PM

Workspace: R's short-term memory

The **workspace** (or global environment) is R's short-term memory: every object you create during the session lives there.

It disappears when R closes, unless you save it.



List what is in memory

`ls ()`



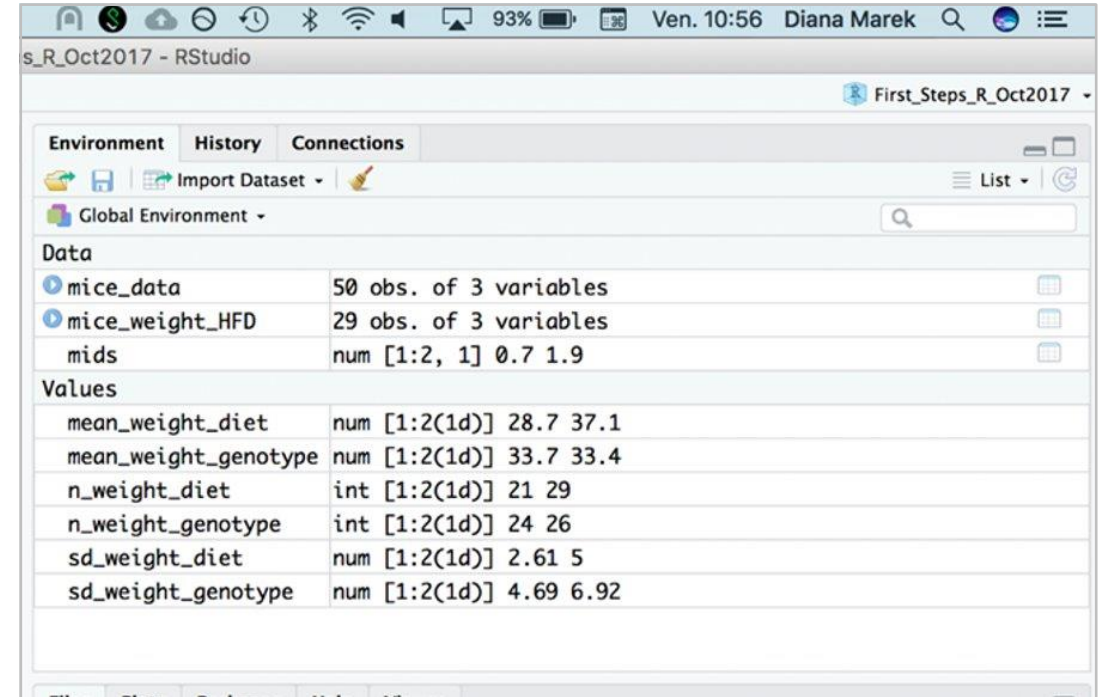
Remove one object

`rm (x)`



Clear everything

`rm(list = ls())`



In RStudio: the Environment pane shows the same objects as `ls()`. The broom icon clears the workspace.

Let's practice - 2

1. Prepare your first script

1. Open a script file and save it.
2. Type or paste the following code

```
# First steps with R, ex1
```

```
w <- 3
```

```
h <- 0.5
```

```
area <- w*h
```

```
area
```

2. **Look at the script** (before running it) can you understand each line? What do you expect it to print to the console?

3. **Run the script** and explore Rstudio's feature:

Run the script line-by-line or Run all lines at once by selecting them

Closing or Switching Projects

- Closing a project: File > Close
- Switch to another project: File > Open Projects...
- Open another project but leave the current one open: File > Open Project in New Session

In your computer's file explorer, **double-clicking a .Rproj file** will open a project

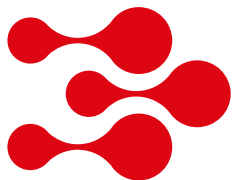
- When closing or switching project, the workspace and history are cleared
(Tools > Global Options > General to change this behavior)

Closing or Switching Projects

- Closing a project: File > Close
- Switch to another project: File > Open Projects...
- Open another project but leave the current one open: File > Open Project in New Session

In your computer's file explorer, **double-clicking a .Rproj file** will open a project

- When closing or switching project, the workspace and history are cleared
(Tools > Global Options > General to change this behavior)



history and workspace can be saved in an .Rhistory and .Rdata file

Projects in Positron: just folders

Positron has no .Rproj file: a project is simply a **folder** (called a workspace). The folder you open becomes the working directory.

	RStudio	Positron
Create	File > New Project	File > New Folder (from a template)
Open	File > Open Project, or double-click the .Rproj file	File > Open Folder, or right-click a folder > Open with Positron (Windows)
Switch	File > Open Projects...	File > Open Recent
Two at once	Open Project in New Session	Open the other folder in a new window
Settings	stored in the .Rproj file	stored in .vscode/settings.json

One folder, both IDEs

Rcourse/

```
| data/
| scripts/
|   └─ ex1.R
| Rcourse.Rproj RStudio
└─ .vscode/ Positron
    └─ settings.json
```

Positron ignores the .Rproj file, so both IDEs can open the same folder.

Same rule as in RStudio: one folder per analysis, open the folder, and use relative paths inside it.

Let's practice - 2bis

1. Look at your project option (Tools > Project Options). If needed, modify them to save your workspace and history and to restore them at startup.
2. Check that it works:
 1. Close Rstudio
 2. Double-click the .Rproj file
 3. Does the project open? Is your workspace empty?
3. Check other behaviours:
 1. Close your project
 2. Open your project again from Rstudio

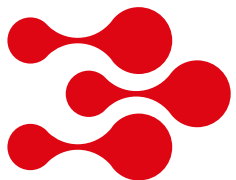
Saving the Workspace (.Rdata file)

You have the option of saving one or several variable to a file for later use

```
save.image("my_workspace.RData") # saves the entire workspace  
save(x,y,file="precious_objects.RData") # saves specific objects
```

You can use `load()` to read the objects in any R session:

```
load("precious_objects.RData")
```



**This is useful to save intermediate results files
Or to pass a bunch of R objects to someone else**

Packages

A **package** is a bundle of R code that someone wrote, documented and shared, so that others can reuse it.

- Installing a package adds new functions (and sometimes datasets) to R.
- Base R comes with only a small set of packages; thousands more are developed by the community.

Analogy: R is a smartphone, packages are its apps. You install an app once, and open it each time you want to use it.

Two main repositories for life sciences

- **CRAN** (cran.r-project.org): the main R repository
- **Bioconductor** (bioconductor.org): bioinformatics, -omics analysis. Installed with `BiocManager::install()`, not `install.packages()`

Inside a package, e.g. `readxl`

Functions

`read_excel()`, `excel_sheets()`

Documentation

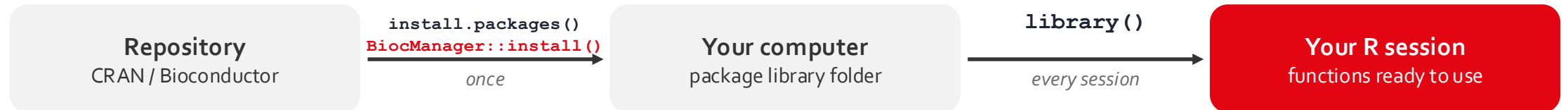
?`read_excel`, examples, tutorials

Example data

small files to try the functions

Dependencies

other packages it needs to work



Installing / loading packages

```
install.packages("stringi") # looks for and installs the package  
named stringi
```

Installation is done only once per computer (or if you reinstall/upgrade R version)

```
library(stringi) # loads library stringi  
#               -> makes its content available in the R session
```

Loading is done at each session, usually at the beginning of a script

Session information

- Getting precise information about your session can be useful for debugging
- This information is usually required to get help online
- `R.version.string` : prints the currently used R version
- `sessionInfo()` : prints version information about R and loaded libraries

Reproducible environnements with Renv

The Problem

"It works on my machine..."

R packages update constantly.

The same code can give **different results** depending on **when** and **where** you run it.

Package versions change → analyses break
→ results differ between collaborators

The Solution

renv records the exact version of every package your project uses.

Initialise a new project

```
renv::init()
```

Freeze current package versions

```
renv::snapshot()
```

Restore environment on any machine

```
renv::restore()
```

All versions saved in **renv.lock** — commit it with your project

Getting started with the R syntax and Objects

What is a data type, and why do we need it?

Inside the computer, everything is stored as 0's and 1's. **Same digits, different type**

A **data type** tells R how to interpret those bits, and therefore **which operations make sense**.

Why it matters

- Arithmetic only works on numbers
- Text needs other tools (pasting, searching, alphabetical sorting)
- TRUE / FALSE values drive decisions and counting
- A wrong type is a classic bug: a column of numbers read as text because one cell says "n/a"

Check the type of any object with **class(x)**

```
> 12 + 1
[1] 13
> "12" + 1
Error: non-numeric argument
      to binary operator

> sort(c(2, 10, 9))
[1] 2 9 10
> sort(c("2", "10", "9"))
[1] "10" "2"  "9"

> class("12")
[1] "character"
```

As text, "10" comes before "2" (compared character by character)

Basic data types

- **Numeric:**

- Number, stored with decimal point
- Examples: 0, 1, 55.2, -11.1111
- In some context, this type is labelled "double"
- Integers, stored without the decimal points, exist but are rarely used

- **Character:**

- Text sequence. Must be enclosed in quotes : " " or ' ' (either work)
- Examples: "1a++", 'Hello World', "s" , "99"

- **Logical:**

- TRUE or FALSE (all upper case)

R syntax

- Case sensitive: R differentiate lower and upper case letters
- Commands can be separated by a newline or a semicolon ; (but newline are preferred for readability)
- Long statements can be written on several lines
- R has no strict rules about the number of spaces around elements. Use your best judgment and be consistent.
- The # character stands for **comments**

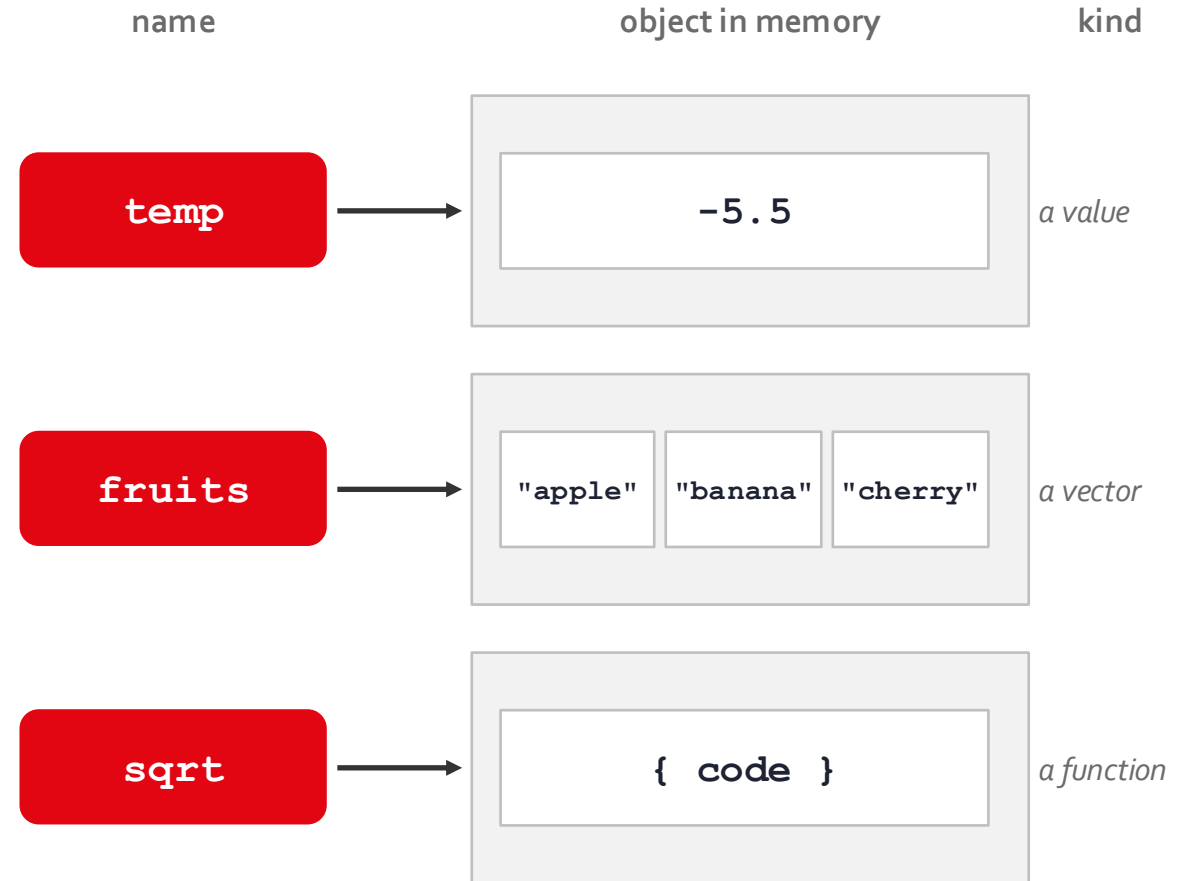
R objects

An **object** is a storage space in memory that contains a value, a data structure, or some code.

Almost everything in R is an object!

- Variables are objects containing data
- Functions are objects containing code

*Think of a labelled box: the **name** is the label you use to find it, the **object** is what is stored inside.*



`ls ()` lists the names of all objects in your workspace

Object name rules

- Letters, numbers, dots . , and underscores _
- Cannot start with a number
- Cannot contain operators
- Best to start with a letter

Examples:

- X
- mydata1
- mydata.normalized
- N_times

The assignment operator

Assigning means storing a value in memory and giving it a **name**, so you can reuse it later.

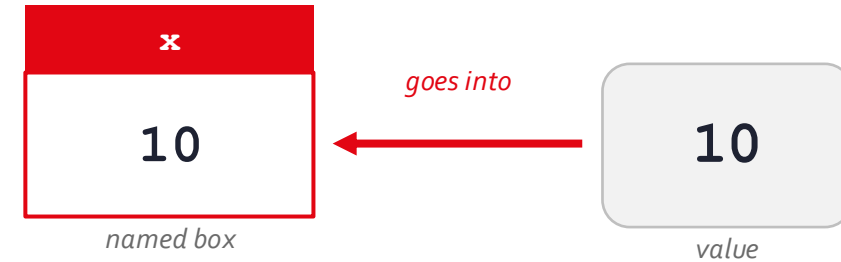
- R first computes the right-hand side, then stores the result under the name on the left.
- Assignment prints nothing: type the name to see the value.
- Either `<-` or `=` works. Pick one and stick to it (`<-` is the R convention).

```
x <- 10           # create x, store 10 in it
x <- 25           # replace the value of x by 25

myNumber <- 15
x <- myNumber     # both now contain 15

x <- x + sqrt(16) # add sqrt(16) to x
```

x <- 10



x <- x + sqrt(16)

① **right side first:** x is still 10, so

10 + 4 → 14

② **then store:** 14 goes into x
the old value 10 is replaced

So this is not an equation, it is an action.

What is a function?



A **function** is a named, reusable piece of code: it takes **inputs**, performs a task, and **returns a result**.

Same idea as in maths: $f(x) = \sqrt{x}$, so $f(81) = 9$

Why use functions?

- Write once, reuse many times, instead of copy-pasting
- Hide complexity: you can use `mean()` without knowing how it is computed
- Already tested by thousands of users: fewer errors

Anatomy of a function call

`round(3.14159, digits = 2)`

function name

1st argument

named argument

`[1] 3.14` ← the returned value

On day 2 you will write your own functions.

Using functions (I)

- Functions are called with parentheses () after the function name
- Arguments are the inputs to functions, passed inside the ()

```
ls()          # no arguments
```

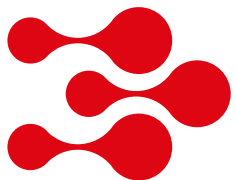
```
sqrt(81)     # one argument
```

```
rep(1, 2)    # two arguments, separated by a comma ,
```

- Arguments have names which can be used when calling a function

```
rep(x=1, times=5) # named args. Equivalent to rep(1, 5)
```

```
rep(1, times=5)   # mixed unnamed/named
```



Use ?function_name to check R's help and see which arguments a function expects

Using functions (II)

Many functions take more than one argument

- If unnamed, arguments must be listed in the correct order (association by position)
- If named, arguments can be passed in arbitrary order (association by name)

```
write.table(object, "outfile.txt" , TRUE)
```

```
write.table(object, append=TRUE, file="outfile.txt")
```

Using functions (III)

Some functions have arguments with default values

Example: **round()**

Usage (from R help): `round(x, digits = 0)`

- Arguments with default values can be omitted in the function call.
- Arguments without default values cannot be omitted.

```
round(2.011)          # rounding to 0 digits, the default  
[1] 2
```

```
round(2.011, 2)       # rounding to 2 digits after the decimal point  
[1] 2.01
```

What is a loop, and why do we need it?

Without a loop: copy, paste, repeat

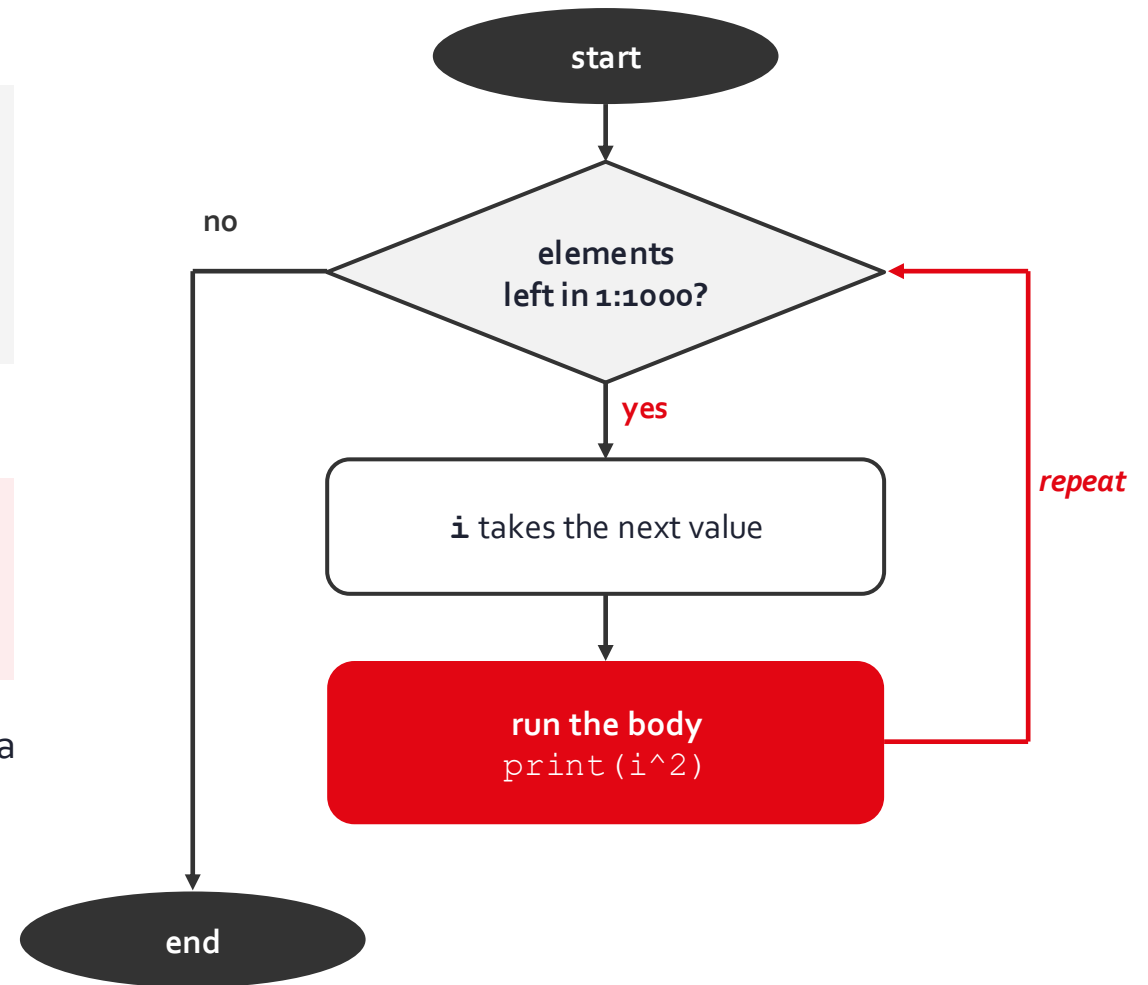
```
print(1^2)
print(2^2)
print(3^2)
# ... and 997 more lines?
```

With a loop: write the action once

```
for (i in 1:1000) {
  print(i^2)
}
```

A **loop** repeats the same block of code once for each element of a collection.

Why? Process many samples, files or genes with the same code: less typing, no copy-paste errors, and it scales from 3 to 3000.



Control flow (I): for loops

Looping through elements

```
for (variable in collection) { body }
```

variable takes each value in turn **collection** is what we go through **body** is the code repeated each time

Example 1: over numbers

```
for (i in 1:5) {  
  print(i^2)  
}
```

Console output

```
[1] 1  
[1] 4  
[1] 9  
[1] 16  
[1] 25
```

Example 2: over a vector of text

```
fruits <- c("apple", "banana", "cherry")  
for (f in fruits) {  
  cat("I like", f, "\n")  
}
```

Console output

```
I like apple  
I like banana  
I like cherry
```

Control flow (II): making a decision with if

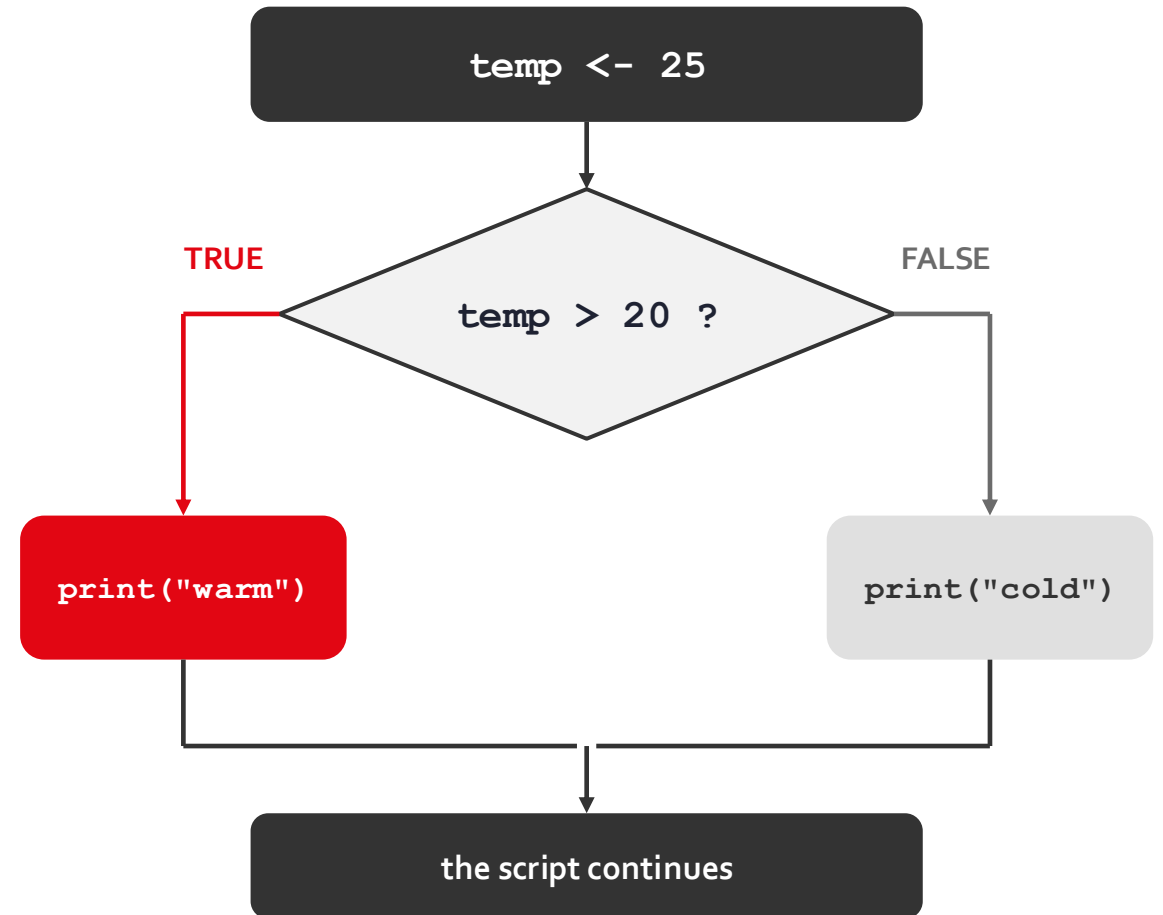
if lets your code **make a decision**: run a block of code only when a condition is **TRUE**, and optionally another block (**else**) when it is **FALSE**.

```
if (condition) {  
  code if TRUE  
} else {  
  code if FALSE  
}
```

Example

```
temp <- 25  
if (temp > 20) {  
  print("warm")  
} else {  
  print("cold")  
}
```

```
[1] "warm"
```



`temp > 20` is a comparison: it returns one **TRUE** or **FALSE**. **else** is optional.

Control flow (IV): loops + conditions

Combine **for** and **if** / **else**: label each temperature and keep the result

```
temps <- c(12, 25, 31, 8)
labels <- character(length(temps)) # empty box

for (i in seq_along(temps)) { # i = 1, 2, 3, 4
  if (temps[i] > 30) {
    labels[i] <- "hot"
  } else if (temps[i] > 15) {
    labels[i] <- "mild"
  } else {
    labels[i] <- "cold"
  }
}

labels
```

```
[1] "cold" "mild" "hot"  "cold"
```

What happens at each turn of the loop

i	temps[i]	condition met	labels[i]
1	12	none	"cold"
2	25	> 15	"mild"
3	31	> 30	"hot"
4	8	none	"cold"

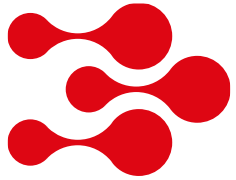
Same result without a loop

`ifelse()` works on the whole vector at once:

```
ifelse(temps > 30, "hot",
      ifelse(temps > 15, "mild", "cold"))
```

Question: which label does 30 °C get? And 15 °C?

Using functions (IV): getting help



Using and understand the help/documentation is 50% of what makes a programmer

`?paste`

`?round`

`?sqrt`

Also, internet is your friend:

- Google "R how to compute square root"
- <https://stackoverflow.com/questions/tagged/r>

Let's practice - 3

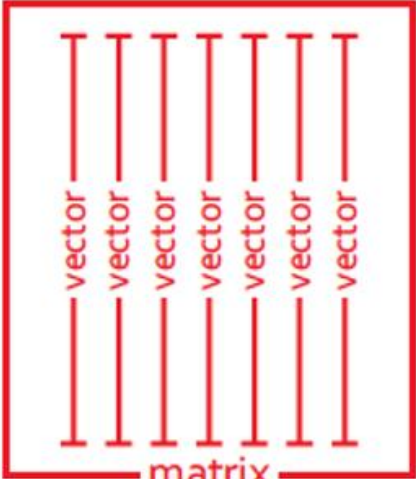
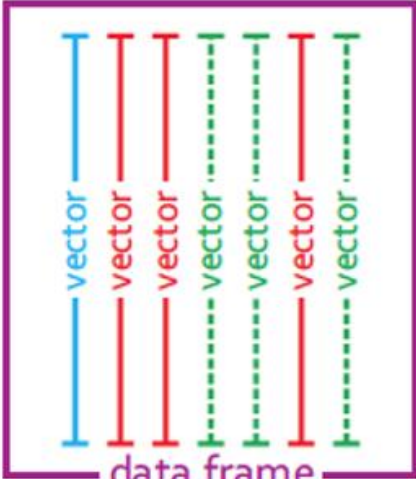
Open a new script and save it

1. Assign the values 6.7 and 56.3 to variable **a** and **b**, respectively
2. Calculate $(2*a)/b+(a*b)$ and assign the result to variable **x**. Display the content of
3. Find out how to compute the square root of variables. Compute the square root of **a**, of **b**, and of **a/b**
4. Calculate:
 - a. the logarithm of **x**
 - b. the logarithm in base 2 of **x**
5. Print the result of a function of your choice (e.g. `sqrt()`) applied to **a** and **b**, using a for loop.
Hint: group the two values with `c(a, b)` (this is called a vector, more on that later), then loop over it
6. Extra task: modify your loop so that it prints a different result depending on the value, e.g. the logarithm if the value is greater than 10, the square root otherwise.
Hint: use `if (...) { ... } else { ... }` inside the loop.
7. Extra task: look into `while()`. How is it different from `for()` ?

Common Object Classes

- **vector**: a series of data, all of the same type
- **matrix**: multiple columns of the same length and data type
- **data frame**: multiple columns of the same length, but each with their data type
- **list**: collection of objects; can be of different classes and sizes
- **function**: a command to perform a specific task

Common Object Classes

				
dimension	1	n	2	1
element data type	single	single	multiple	multiple
element data structure	atomic	atomic	vector	any
subsetting	x[i] x["name"] x[1:3]	x[i,j,...] x["row","col",...] x[,1:3,...]	x[i,j] x["row","col"] x\$colname	x[[i]] x\$colname

Vectors: simple usage

Create a vector using `c()` :

```
height <- c(180, 167, 199) # c() is for concatenate
```

Create a named vector:

```
height <- c(Mia=180, Paul=167, Ed=199)
```

Access elements with `[]`:

```
height[1] # get the first element
```

```
height["Paul"] # get the element named Paul
```

```
height[ c(1,3) ] # get the 1st and 3rd element
```

Vectors: more creation recipes

```
a <- 1:10          # the : operator
  1 2 3 4 5 6 7 8 9 10
```

```
s <- seq(0,2,0.5)  # seq: from 0 to 2 by increment of 0.5
  0.0 0.5 1.0 1.5 2.0
```

```
genotypes <- rep( c('WT','KO') , 3 ) # rep to repeat
  "WT" "KO" "WT" "KO" "WT" "KO"
```

```
c(a,s)            # c() can be used to concatenate 2 vectors
  1.0  2.0  3.0  4.0  5.0  6.0  7.0  8.0  9.0
10.0  0.0  0.5  1.0  1.5  2.0
```

Vectors: manipulation (I)

```
a <- 1:4
```

```
a
```

```
[1] 1 2 3 4
```

```
a*2 # multiply each element of a by 2
```

```
[1] 2 4 6 8
```

```
a + c(12,10,12,10) # add the elements of 2 vectors
```

```
[1] 13 12 15 14
```

Vectors: manipulation (II)

Many functions take a vector as argument.

Some perform an element-wise operation. Example:

```
log2(a) # compute the logarithm in base 2 of each element
```

```
[1] 0.000000 1.000000 1.584963 2.000000
```

Some return a single value. Example:

```
mean(a) # compute the mean of the elements of a
```

```
[1] 2.5
```

Coercion (I)

- All elements of a vector must be the **same type**
- When combining different types, they are **coerced to the most flexible type**
- Flexibility scale: logical < numeric < character

```
vec <- c(12, "twelve", TRUE) # combine 3 data type
vec                          # coerced to character
[1] "12" "twelve" "TRUE"
```

```
class(vec) # class() returns the type of a vector
[1] "character"
```

Coercion (II)

We can coerce a vector to any type with: `as.logical()` , `as.numeric()`, `as.character()`

Example: coerce a logical vector to numeric

- FALSE becomes 0, TRUE becomes 1
- mathematical functions automatically coerce to numeric

```
x <- c(FALSE, FALSE, TRUE)
```

```
as.numeric(x) [1] 0 0 1
```

```
sum(x)      # number of TRUE
```

```
mean(x)     # proportion of TRUE
```

Let's practice - 4

1. Create two vectors:
 - **vector_a**, containing the values from -5 to 5
 - **vector_b**, from 0 to 1 by increment of 0.1
2. Calculate the (element-wise) sum, difference and product between the elements of **vector_a** and **vector_b**.
3. Calculate the sum of elements in **vector_a**.
4. Calculate the overall sum of elements in both **vector_a** and **vector_b**.
5. Identify the smallest and the largest value in **vector_a**
6. Identify the smallest and the largest value among both **vector_a** and **vector_b**.
7. Compute the overall mean of the values among both **vector_a** and **vector_b**.

*Hint: Each task in exercises 1-7 can be performed in a single statement per vector
(the minimum and maximum count as 2 separate tasks)*

EXTRA TASKS:

- use the `seq()` function to create a vector **x** containing 100 values equally spread between 0 and 5
- compute a vector **y** corresponding to the exponential of **x**
- compute and compare the means and standard deviations of **x** and **y**

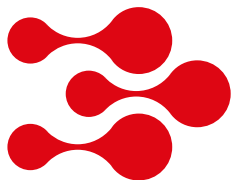
Operators (most common ones)

Arithmetic: `+` , `-` , `*` , `/` , `^`

Comparison: `<` , `>` , `<=` , `>=` , `==` (equal to),
`!=` (different from)

Logical: `!` (negation) , `&` (and) , `|` (or)

Other: `%in%`



Comparison, logical operators, and `%in%` always return logical values (TRUE/FALSE)

Operators returning logical values

```
c(1, 3, 2) == 2
```

```
[1] FALSE FALSE TRUE
```

```
c(1, 3, 2) < 2
```

```
[1] TRUE FALSE FALSE
```

```
!(c(1, 3, 2) < 2)    # ! Reverses TRUE and FALSE
```

```
[1] FALSE TRUE TRUE
```

```
c("Fred", "Marc", "Dan", "Ali") %in% c("Dan", "Geoff", "Ali")
```

```
[1] FALSE FALSE TRUE TRUE
```

Using logical vectors to subset

The `[]` operator can also accept a logical vector

```
a <- 1:4
```

```
a
```

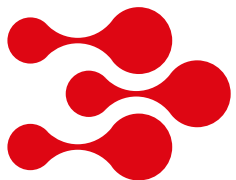
```
[1] 1 2 3 4
```

```
a[ c(FALSE, FALSE, TRUE, TRUE) ] # select the elements where there is TRUE
```

```
[1] 3 4
```

```
a[ a>2 ] # the logical vector is provided by a logical expression
```

```
[1] 3 4
```



**This sort of operation is extremely common
to subset some data**

Missing values (I)

Missing values are usually represented by NA:

```
y <- c(1, 2, 3, 4, 5, NA, NA)
```

NA's interfere with many functions:

```
mean(y)  
[1] NA
```

Arguments often exist to remove NA's before calculation

```
mean(y, na.rm=TRUE)  
[1] 3
```

Alternatively, use **na.omit()** to remove NAs from the data

```
y_cleaned <- na.omit(y)  
mean(y_cleaned)  
[1] 3
```

Missing values (II)

```
x <- c(1, NA, 0/0) # 0/0 gives NaN = Not a Number
```

```
x
```

```
x [1] 1 NA NaN
```

```
is.na(x) #detects NAs and NaNs from x
```

```
[1] FALSE TRUE TRUE
```

```
is.nan(x) # detects only NaNs from x
```

```
[1] FALSE FALSE TRUE
```

```
x > 2 # what if we try to compare NA and NaN to a number?
```

```
[1] FALSE NA NA
```

```
x[!is.na(x)] # removes NAs and NaNs from x
```

```
[1] 1
```

Data frame: creation

We can create a dataframe from its individual column (each one is a vector)

```
name          <- c("Joyce", "Chaucer", "Homer")
status        <- c("dead", "deader", "deadeast")
reader_rating <- c(55, 22, 100)
```

```
poets <- data.frame(name, status, reader_rating)
poets
```

	name	status	reader_rating
1	Joyce	dead	55
2	Chaucer	deader	22
3	Homer	deadeast	100

Data frame: manipulation

```
poets[ 2 , 2 ]      # gets the element on row 2 in column 2
poets[ , c(1,3) ]   # gets columns 1 and 3
poets[ , c("name", "reader_rating") ] # gets columns "name" and
                                     # "reader_rating"
poets$name          # gets column "name"

rownames(poets)     # gets the row names
colnames(poets)     # gets the column names

rownames(poets) <- c("J", "C", "H") # overwrites row names
```

Let's practice - 5

Open a new script and save it as "Ex5.R".

1. In your script, write and execute the commands

```
library(MASS) # loads the library MASS
data(bacteria) # loads the bacteria data set (from MASS)
```

Check: You should have a variable named "bacteria" in your Environment.

2. What are the names of the columns of the **bacteria** data.frame ?
3. Use `[]` to select rows 100 to 119 of the column `ap`.
4. Use `$` to get the column `week` and check how many 0 values it has.

EXTRA TASKS

1. using a comparison operator and `[]`, select the rows which correspond to a "placebo" treatment (in the `trt` column).
2. Compute the fraction of "y" in columns `y` (*hint: the function `mean()` can help*).
3. Compute the same fraction but only among the rows corresponding to a "placebo" treatment. Same with a "drug" treatment, and with a "drug+" treatment.

Lists: creation

Lists are collections of object which may be of different classes and sizes

Create a few objects:

```
vec <- c(0.4, 0.9, 0.6)
```

```
mat <- cbind(c(1,1), c(2,1))
```

```
df <- data.frame(name=c("Ed", "Lisa"), age=c(61, 71))
```

Unnamed list - collect these objects in a list, using the function **list()**:

```
l <- list(vec, mat, df)
```

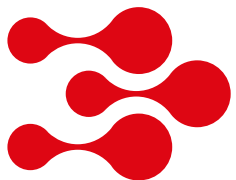
Named list - collect these objects in a list with named elements:

```
l_with_names <- list(myvector=vec, mymatrix=mat, mydata=df)
```

Lists: manipulation

```
l[[1]]           # gets the first object
l_with_names[["myvec"]] # gets the object named "myvec"
l_with_names$myvec  # gets the object named "myvec"

names(l_with_names) # gets the list elements' names
names(l_with_names) <- c("A", "B", "C") # overwrites names
```



Lists can be used to collect a diverse sets of objects related to the same analysis

Everything in R is an object

Using R is all about creating and manipulating data objects using functions

Objects have a class (vector, data frame, list,...)

Data in objects have a type (numeric, character, logical)

Formatting your Data

Prepare your Data outside of R

To make importation of your data in R easy, you have to format it properly beforehand.

Excel or OpenOffice can be used for that task.

Three main precepts of tidy data:

- Each variable forms a column
- Each observation forms a row
- Each type of observational unit forms a table

<https://www.ucd.ie/ecomodel/resources/importingdataintor/>

Example of a well-formatted dataset

	A	B	C	D
1	chr	pos	minor	major
2	1	123369	A	C
3	1	138369	G	T
4	1	153369	T	C
5	1	168369	C	T
6	1	183369	G	A
7	1	198369	T	A
8	5	228369	G	A
9	5	258369	G	A
10	5	288369	A	G
11	5	318369	C	A
12	5	348369	A	T

- A header line with variable names
- 4 variables, 1 in each column
- One observation per row

Example of a well-formatted dataset - metadata

	A	B	C	D
1	chr	pos	minor	major
2	1	123369	A	C
3	1	138369	G	T
4	1	153369	T	C
5	1	168369	C	T
6	1	183369	G	A
7	1	198369	T	A
8	5	228369	G	A
9	5	258369	G	A
10	5	288369	A	G
11	5	318369	C	A
12	5	348369	A	T

Glossary for the snp data

snp single nucleotide polymorphism

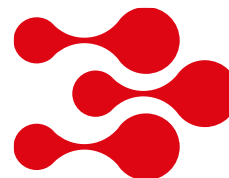
chr chromosome

pos position

minor minor allele (a minority of individuals
have this letter at this position)

major major allele (a majority of individuals
have this letter at this position)

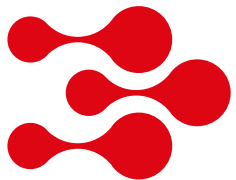
+ data source, date of collection,...



**Data is often useless without the
accompanying metadata**

But the reality is not always as simple...

Two exercise: A) metabolites_table and optional: B)
Cell_lines_table



**Difficult to use with any
Statistical program**

Courtesy of Frederic Schuetz

Importing data

Most flexible function to read tabular data: `read.table()`

- Reads a formatted text file
- Imports it as a **data frame**
- **Many** options, to accommodate most text files

Two main elements to give to `read.table()`:

- What/where is the file to read
- What are the formatting options

Specifying the file to read (I)

Where is the file I want to import?

- Look for your file in the file system.
- Note its **path**: the succession of folders to access it

File paths can be specified as a string with '/' as separator:

```
"C:/Users/Leo/courses/data/snp.csv"
```

Or with a little help from the function **file.path()**:

```
file.path("C:", "Users", "Leo", "courses", "data", "snp.csv")
```

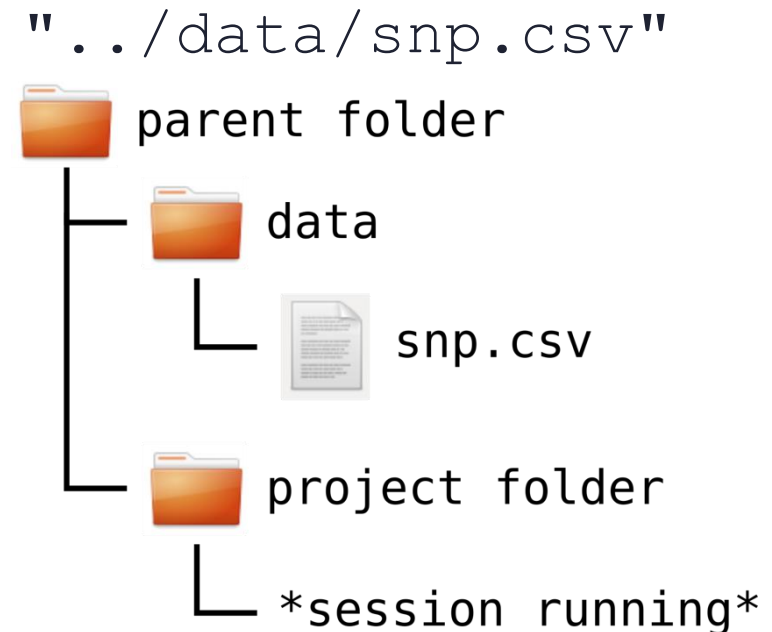
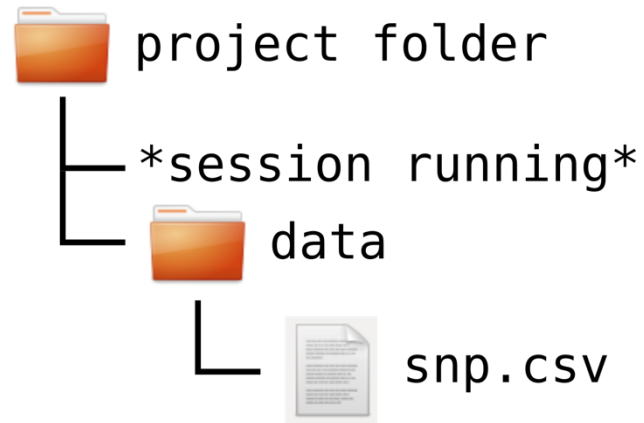
Specifying the file to read (II)

Files path can be **absolute**

```
"C:/Users/Leo/courses/data/snp.csv"
```

Or **relative** (to the working directory)

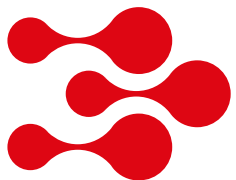
```
"data/snp.csv"
```



Format options

Most important optional arguments of `read.table()`, `read.csv()`, `read.delim()`

- **header** (TRUE/FALSE): specifies whether the first line contains column names
 - Default in `read.table()` is FALSE.
 - Default in `read.csv()` and `read.delim()` is TRUE.
- **sep**: specifies the field separator character (e.g. "," or tab "\t").
 - Default in `read.table()` is any white space characters (space, tab, newline and carriage return).
 - Default in `read.csv()` is comma.
 - Default in `read.delim()` is tab.



When in doubt, use `help(read.table)`

Importing data

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File paths can be specified as a string with '/' as separator:

```
"C:/Users/Leo/courses/data/snp.csv"
```

Or with a little help from the function **file.path()**:

```
file.path("C:", "Users", "Leo", "courses", "data", "snp.csv")
```

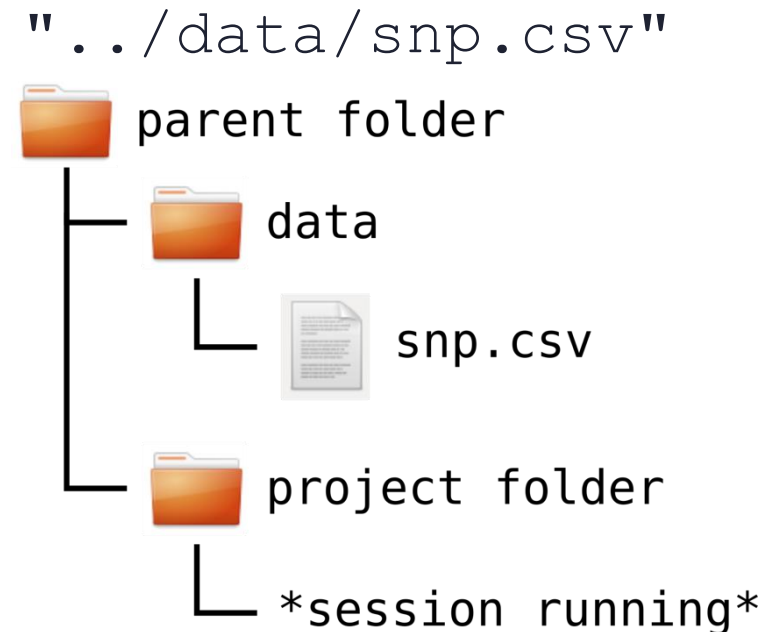
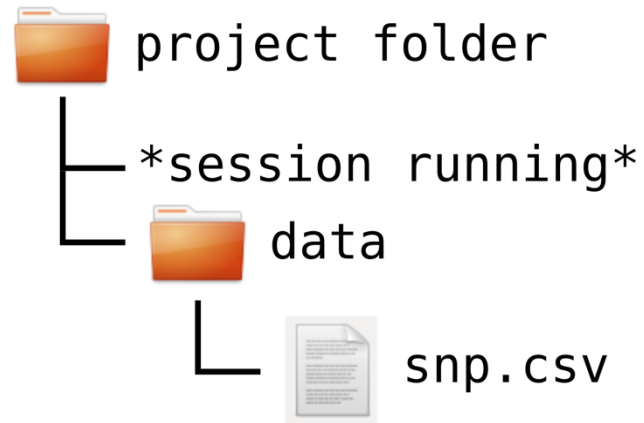
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Files path can be **absolute**

```
"C:/Users/Leo/courses/data/snp.csv"
```

Or **relative** (to the working directory)

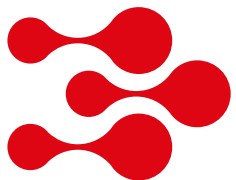
```
"data/snp.csv"
```



Format options

Most important optional arguments of `read.table()`, `read.csv()`, `read.delim()`

- **header** (TRUE/FALSE): specifies whether the first line contains column names
 - Default in `read.table()` is FALSE.
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 - Default in `read.delim()` is tab.



When in doubt, use `help(read.table)`

Reading Excel files

Although it is recommended to export your data files to a non-proprietary text format, it is still possible to read directly from an excel file.

```
install.packages("readxl")
```

```
# Loading
```

```
library(readxl)
```

```
# reading the sheet named 'my data' from an excel file
```

```
my_data <- read_excel("my_file.xls", sheet='my data')
```

Metabolites dataset (from Bosnjakovic et al., BioRxiv, 2025)

doi:<https://doi.org/10.1101/2025.07.10.663939>

Bosnjakovic2025_metabolites_dataset.xlsx

Look at the table in R studio's file explore. Then try to read it into R. Remember we talked about `read.delim`, `read.table`, `read.xlsx`...

➤ What do you observe?

Metabolites dataset

Bosnjakovic2025_metabolites_dataset.xlsx

Row 1 : treatment → None None None None None "VC, BaP" "VC, BaP" "BaP"
...

Row 2 : condition → None None None None None "0.2% DMSO" "0.2%
DMSO" "20μM BaP" ...

Row 3 : real names → "Precursor Name" "Prec. Form." ... "24h" "24h" "24h" ...

Step-by-step

If you want to use base R `read.delim` or `read.table`, you need to:

- 1) Open the excel file
- 2) Save the sheet you are interested in as tab delimited txt file
- 3) Read it in R

```
# Reading and formatting a metabolomics table from Excel

#### with read.delim() or read.table() you need a txt file format

file <- "Bosnjakovic2025_metabolites_pos.txt"

# Read only the 3 header rows
#   nrow = 3      -> read at most 3 rows, then stop
#   header = FALSE -> do NOT use the first row as column names
#   colClasses = "character" -> read everything as text
meta <- read.delim(file, header = FALSE, nrow = 3,
                   colClasses = "character", fileEncoding = "UTF-8")

# Read the data
#   skip = 3      -> ignore the first 3 rows, read everything after them
abund <- read.delim(file, header = FALSE, skip = 3, fileEncoding = "UTF-8")

meta  # look at the header rows
str(abund) # check the data: abundances should be numbers (num / int)
```

```
#### Better option?
```

Step-by-step

```
library(readxl)
file <- "Bosnjakovic2025_metabolites_dataset.xlsx"

# 1. Read the 3 header rows and the data separately
meta <- read_excel(file, sheet = "metabolites_pos", n_max = 3, col_names = FALSE)
abund <- read_excel(file, sheet = "metabolites_pos", skip = 3, col_names = FALSE)

# 2. Turn each header row into text
row1 <- as.character(unlist(meta[1, ]))
row2 <- as.character(unlist(meta[2, ]))
row3 <- as.character(unlist(meta[3, ]))

# 3. Empty cells (NA) become ""
row1[is.na(row1)] <- ""
row2[is.na(row2)] <- ""
row3[is.na(row3)] <- ""

# 4. Build unique column names
col_names <- paste(row1, row2, row3, sep = "_") # paste the 3 rows with "_"
col_names <- gsub("^_+_+$", "", col_names)      # remove "_" at start/end
col_names <- make.unique(col_names)             # duplicates get .1, .2
colnames(abund) <- col_names

# 5. MOI sheet: clean the non-breaking spaces
moi <- read_excel(file, sheet = "MOI_identification")
text_cols <- sapply(moi, is.character)
moi[text_cols] <- lapply(moi[text_cols], function(x) {
  x <- gsub("\u00a0", " ", x) # non-breaking space -> normal space
  trimws(gsub("\\s+", " ", x)) # remove extra spaces
})
```

Cell lines dataset (from Ujiie, Sakyo, Oya, et al., Cancer Medicine, 2025)

Doi: <https://doi.org/10.1002/cam4.71373>

Ujiie2025_Supplementary Table S3.xlsx

Again, kook at the table in R studio's file explore, try to read it into R. What do you observe in this case?

Cell lines dataset (from Ujiie, Sakyo, Oya, et al., Cancer Medicine, 2025)

Doi: <https://doi.org/10.1002/cam4.71373>

The problem: line 1 table title, real column names on line 2, NA in *PRISM_IC50* for the excluded cells, column name in *CamelCase*.

Step-by-step

```
library(readxl)

# 1. Read the table, skipping the title row
cell_lines <- read_excel("Ujiie2025_Supplementary Table S3.xlsx",
                        sheet = "Table S3", skip = 1)

# 2. Clean the column names (replaces clean_names())
names(cell_lines) <- gsub("[a-z0-9][A-Z]", "\\1\\2", names(cell_lines)) # camelCase -> camel_Case
names(cell_lines) <- tolower(names(cell_lines))                       # all lowercase
names(cell_lines) <- gsub("[^a-z0-9]+", "_", names(cell_lines))         # spaces, %, () ... -> "_"
names(cell_lines) <- gsub("^_|_$", "", names(cell_lines))              # remove "_" at start/end
names(cell_lines) # check the new names

# 3. Remove the "index" column (a 0-based Python row number, not useful)
cell_lines$index <- NULL

# 4. Turn "status" into a factor with a fixed order
cell_lines$status <- factor(cell_lines$status, levels = c("Included", "Excluded"))

# 5. Count the NAs in each column
colSums(is.na(cell_lines))

# NAs in prism_ic50 are structural : some of the "Excluded" miss PRISM values.
# But why not all? What was the reason to be excluded?
# Sometimes we cannot rely on is.na, we need to know the other metadata
```

[a-z0-9] one lowercase letter or one digit

[A-Z] one capital letter

() "remember this part", so it can be reused in the replacement

So the pattern finds every spot where a lowercase letter or digit is directly followed by a capital letter. Those are the word boundaries in camelCase.

The replacement: "\\1\\2" : what the first () found _ what the second found

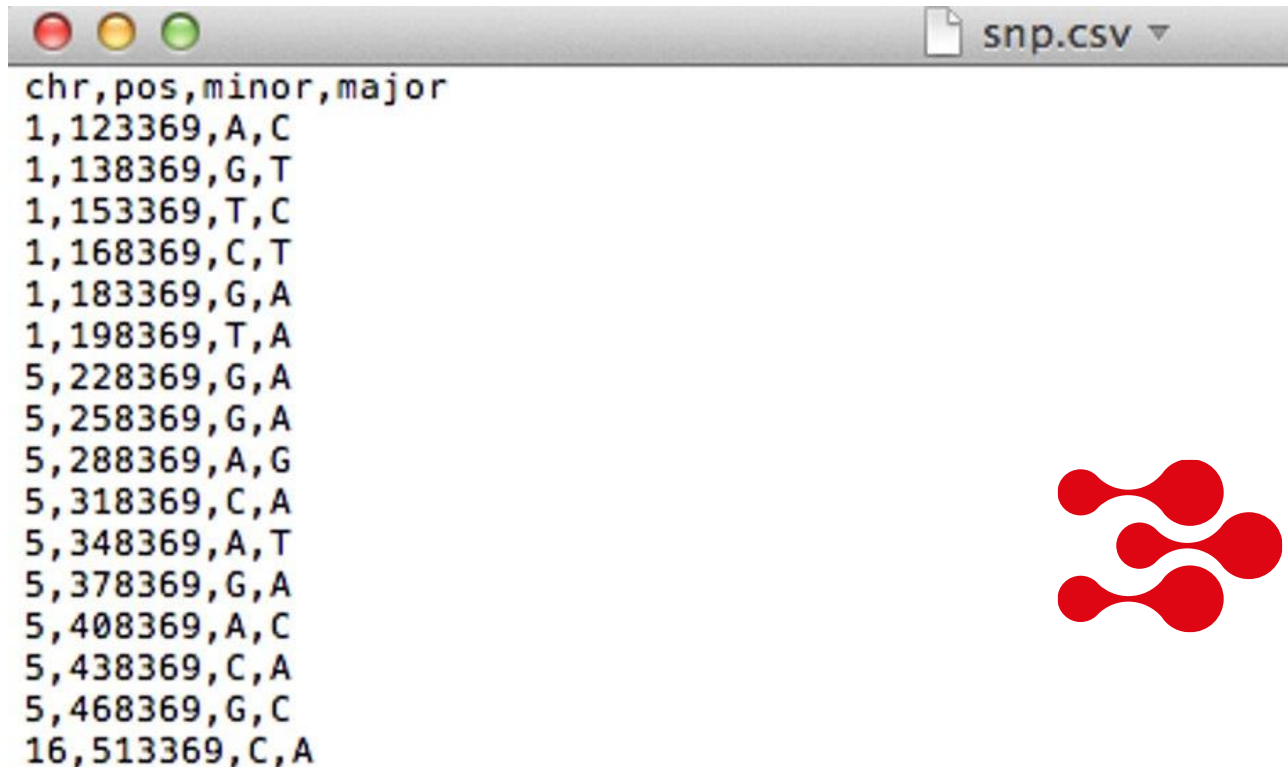
gsub("[^a-z0-9]+", "_", x) # ^ = NOT

gsub("^_|_\$", "", x) # ^ = start of text

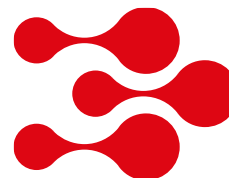
Saving your data

Export the spreadsheet to a text file format

- csv (comma separated values), file extension .csv OR
- tsv (tab separated values), file extension .txt or .tsv



```
chr,pos,minor,major
1,123369,A,C
1,138369,G,T
1,153369,T,C
1,168369,C,T
1,183369,G,A
1,198369,T,A
5,228369,G,A
5,258369,G,A
5,288369,A,G
5,318369,C,A
5,348369,A,T
5,378369,G,A
5,408369,A,C
5,438369,C,A
5,468369,G,C
16,513369,C,A
```



Now you are ready to import this dataset in R

Formatting checklist/recommendations

- First row is a **header** (column names)
- First column contain **row IDs**
- No blank spaces in column names (use _ instead)
- Column names do not contain symbols other than _
- Short column names are better
- Remove all comments or other content around the data table
--> put them in a **metadata file**
- Indicate missing values with **NA** (spelled exactly like that)
- Have data backups!

Let's practice – 6a

A dataset from mouse experiments at 18 weeks is available in the file ***mice_data.csv*** in the ***course_datasets*** folder (courtesy of F. Schutz and F. Preitner). Let's explore the dataset to see what it contains.

1. Open a new script file in R studio, comment it and save it.
2. Have look at the csv file in R studio's file explorer. What do you need to check in order to be able to read in the file correctly?
3. Read the file into R, assign its content to object "mice_data". Examine the object.
4. How many observations and variables does the dataset have?
5. What is the structure of the dataset? What are the names and classes of the variables?

Digression: factors (I)

Factors is a datatype used to represent **categorical data**.

Some functions require factors (and not character) to work properly.

```
genotype <- factor(c("WT", "WT", "Mut2", "Mut1", "Mut2"))  
[1] WT    WT    Mut2  Mut1  Mut2  
Levels: Mut1 Mut2 WT
```

The available values in a factor are called levels.

```
levels(genotype)  
[1] "Mut1" "Mut2" "WT"
```

Digression: factors (II)

By default factor levels are sorted alphabetically.

We can specify them manually (sometimes useful to set reference level in some stat applications)

```
genotype <- factor(c("WT", "WT", "Mut2", "Mut1", "Mut2"),  
                  levels=c("WT", "Mut1", "Mut2"))
```

```
[1] WT    WT    Mut2  Mut1  Mut2
```

```
Levels: WT Mut1 Mut2
```

Setting factor variables

Convert categorical variables to factors as needed.

```
snps$chr    <- factor(snps$chr, levels=c("1","5","16","22"))
snps$minor  <- factor(snps$minor)
snps$major  <- factor(snps$major)
```

```
str(snps)    # structure of the data frame
```

```
'data.frame': 40 obs. of  4 variables:
```

```
$ chr   : Factor w/ 4 levels "1","5","16","22": 1 1 1 ...
```

```
$ pos   : int  123369 138369 153369 168369 183369 ...
```

```
$ minor: Factor w/ 4 levels "A","C","G","T": 1 3 4 2 3 ...
```

```
$ major: Factor w/ 4 levels "A","C","G","T": 2 4 2 4 1 ...
```

Getting a summary

```
summary(snps)
```

chr	pos	minor	major
1 : 6	Min. : 123369	A: 9	A:17
5 : 9	1st Qu.: 340869	C:10	C:10
16: 9	Median : 715869	G:15	G: 4
22:16	Mean : 777869	T: 6	T: 9
	3rd Qu.:1185869		
	Max. :1673369		

Reminder – accessing parts of the dataframe (I)

```
snps[2,] # 2nd row
  chr    pos minor major
2    1 138369     A      T
```

```
snps[, "minor"] # column named "minor"
[1] A A T C G T G G A C A G A C G C A G G C G A C A G T C G C C T A G G T G
T C G A
Levels: A C G T
```

```
snps[1:3, c(1,3)] # 3 first rows, 1st and 3rd column
  chr minor
1    1     A
2    1     G
3    1     T
```

Reminder – accessing parts of the dataframe (II)

```
snps$chr # equivalent to snps[, 1]
[1]  1  1  1  1  1  1  1  5  5  5  5  5  5  5  5  5 16 16
16 16 16 16 16 16 16 22 22 22
[28] 22 22 22 22 22 22 22 22 22 22 22 22 22 22
Levels: 1 5 16 22
```

```
snps$chr[40] # chromosome of the last row
[1] 22
```

table()

The table() function is useful to get a summary of one or several categorical columns.

```
table(snps$chr)
```

```
1  5 16 22  
6  9  9 16
```

```
table(snps$minor, snps$major)
```

```
# rows are minor, columns are major
```

	A	C	G	T
A	0	4	2	3
C	3	0	2	5
G	10	4	0	1
T	4	2	0	0

table()

The table() function is useful to get a summary of one or several categorical columns.

```
table(snps$chr)
```

```
1  5 16 22
6  9  9 16
```

```
table(snps$minor, snps$major , dnn = c("minor","major"))
```

```
# rows are minor, columns are major
```

	major			
minor	A	C	G	T
A	0	4	2	3
C	3	0	2	5
G	10	4	0	1
T	4	2	0	0

Let's practice – 6b

Continue from the mouse dataset used previously.

Use the following code if you do not have the dataframe already loaded

```
mice_data = read.csv("course_datasets/mice_data.csv")
```

1. Which variables appear to be categorical? Convert them to factors.
2. Get the summary statistics of "mice_data"
3. Use the function table() to compute the number of observations in different mouse groups.
 1. How many mice are included of each genotype (WT, KO)?
 2. How many mice are included per diet (HFD, CHOW)?
 3. Make a 2x2 table by genotype and diet crossed.

EXTRA TASKS:

1. Come to the bacteria dataset and explore the relationship between the categorical columns using what you learned

Subsetting (I)

`subset()` allows you to subset your data by specific columns and values in those columns. Logical operators can be used within the subset.

```
subset(snps, chr==1) # keeps only the snps where chr is 1
```

	chr	pos	minor	major
1	1	123369	A	C
2	1	138369	G	T
3	1	153369	T	C
4	1	168369	C	T
5	1	183369	G	A
6	1	198369	T	A

Subsetting (II)

keeps only the snps in chr 1 with an "A" as major allele

```
subset(snps, chr==1 & major=="A")
```

	chr	pos	minor	major
5	1	183369	G	A
6	1	198369	T	A

keeps only the snps in chr 1 with an "A" or "T" as major allele

```
subset(snps, chr==1 & (major=="A" | major=="T"))
```

	chr	pos	minor	major
2	1	138369	G	T
4	1	168369	C	T
5	1	183369	G	A
6	1	198369	T	A

Customizing summaries of data (I)

`tapply()` generates **custom summaries** of your data using :

- **X**: a column you want to aggregate (of any data type)
- **INDEX**: a factor column, or list of factor columns, for grouping
- **FUN**: a function to be applied to X (mean, sd, min, max, length, median, range, quantiles...), separately for each grouping indicated by INDEX

Customizing summaries of data (II)

```
# loading a dataset with hours of extra sleep in two groups
```

```
data(sleep)
```

```
head(sleep, n=3)
```

	extra	group	ID
1	0.7	1	1
2	-1.6	1	2
3	-0.2	1	3

```
tapply(X=sleep$extra, INDEX=sleep$group, FUN=mean)
```

1	2
0.75	2.33

For each **group**, compute the **mean extra** sleep

Adding a row or column

Add a row to the snp data: `rbind()`

```
snps_updated <- rbind(snps,  
                      data.frame(chr=22,  
                                pos=1723369,  
                                minor="A",  
                                major="T"))
```

Add a column to the snp data: `cbind()`

```
majorGC <- snps$major %in% c("G", "C")  
snps_mod <- cbind(snps, majorGC)
```

OR

```
snps$majorGC <- snps$major %in% c("G", "C")
```

Removing a row or column

Exclusion (-):

```
snps_orig <- snps_mod[, -1] # remove the first column  
head(snps_orig) # check resulting data
```

Extraction:

```
# extract all columns that you want to keep  
# (from the 2nd to the last)  
snps_orig <- snps_mod[, 2:ncol(snps_mod)]  
head(snps_orig) # check resulting data
```

Exporting data to a file

`write.table()` or `write.csv()`

Example:

```
write.table(snps_updated, file="snps_updated.csv",  
           quote=FALSE, sep="," , row.names=FALSE)
```

Some important arguments (check ?write.table for more):

- **file:** file path and name for the output file
- **append:** allows to append to an existing file (default is FALSE).
- **quote:** whether the elements of character or factor columns should be surrounded by double quotes in the printed output (default is TRUE).
- **sep:** field separator to be used, e.g., comma (",") or tab ("\t").
- **row.names:** whether the row names are written (default is TRUE)
- **col.names:** whether the column names are written (default is TRUE).

Let's practice – 6c

Continue from the mouse dataset used previously.

Use the following code if you do not have the dataframe already loaded

```
mice_data = read.csv("course_datasets/mice_data.csv")
```

1. Subsets

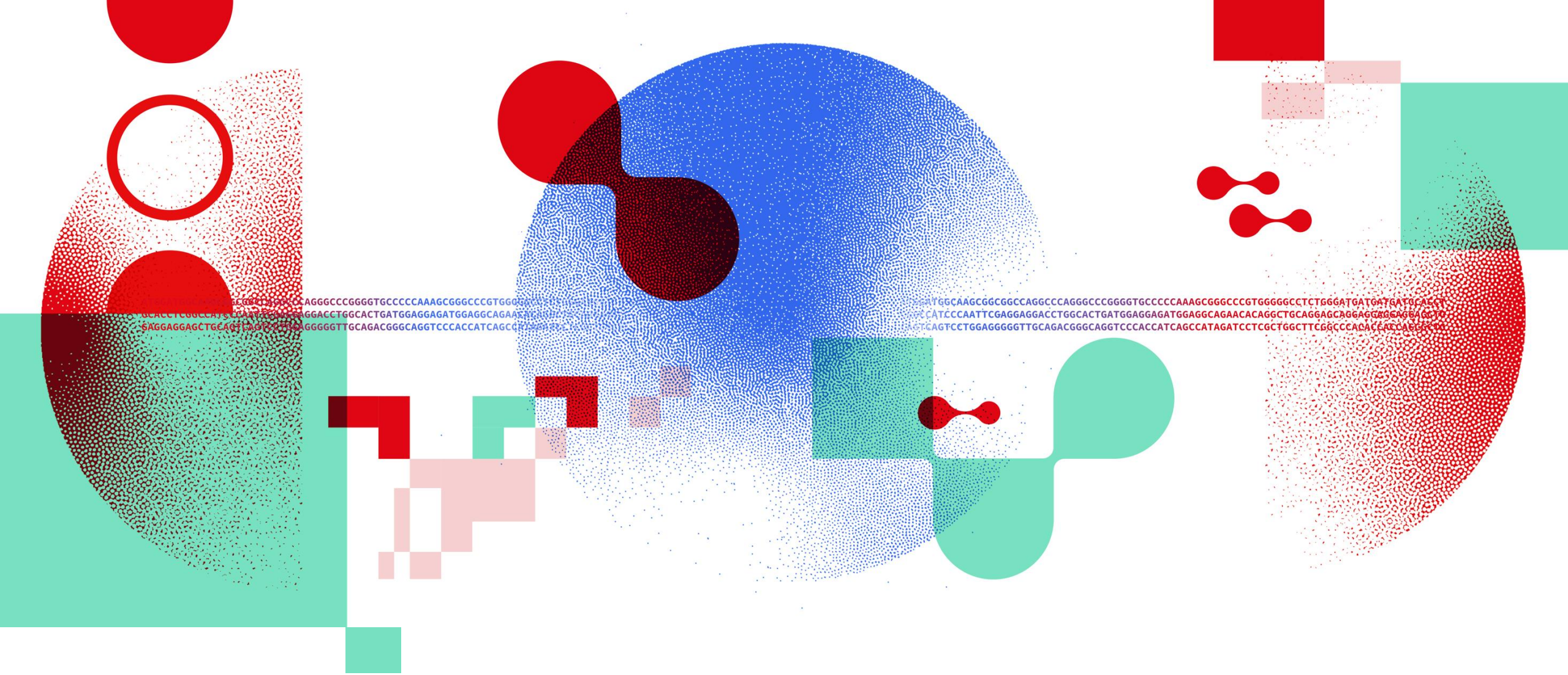
1. Isolate the observations for the mice on high fat diet (HFD) using `subset()`.
 2. Compute the average weights of the subset.
 3. Do the same for the mice on regular chow diet (CHOW).
 4. Export the data of each subgroup to a csv file.
2. Look at the results from the two previous exercises. What does this initial exploration of the data suggest about mouse weights?
 3. **Optional:** Compute the means and standard deviations for WT and KO mouse weights using `tapply()`. Then do the same for CHOW and HFD groups.

EXTRA TASK: next slide

Let's practice – 6c – extra tasks

Come back to the bacteria dataset. We will try to explore how the fraction of infection of *H.influenzae* in otitis changes with respect to both week of observation and treatment

1. add a new column `y_yes` to the dataframe which contains `TRUE` when column `y` is "y" and `FALSE` otherwise. This column will be useful for our `tapply` after.
2. Use `tapply` to compute the fraction of otitis with *H.influenzae* (column `y_yes`) for each week
3. for each level of the `trt` column
 1. Generate a subset of the dataframe with only the observations with this treatment level
 2. Use `tapply` to compute the fraction of otitis with *H.influenzae* for each week for this treatment level



Thank you

Annexes

Reading Excel files

Although it is recommended to export your data files to a non-proprietary text format, it is still possible to read directly from an excel file.

```
install.packages("readxl")
```

```
# Loading
```

```
library(readxl)
```

```
# reading the sheet named 'my data' from an excel file
```

```
my_data <- read_excel("my_file.xls", sheet='my data')
```

Coding style

Different authorities have different style recommendations for naming things, spacing, operator symbols, layout, commenting etc.

Example: <https://web.stanford.edu/class/cs109l/unrestricted/resources/google-style.html>

File names: Use meaningful names, ending with file extension .R (`predict_ad_revenue.R`)

Identifiers: Variable names should have all lower case letters, words separated with dots (`avg.clicks`)

Line length: maximum 80 characters

Indentation: two spaces, no tabs

Assignment: use `<-`, not `=`

Semicolon: don't use them

Spaces:

- Place spaces around all binary operators (`=`, `+`, `-`, `<`, etc.)
- Do not place a space before a comma, but always place one after a comma.
- Otherwise, do not place spaces around code in parentheses or square brackets